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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY**



Embedded System

Mini environmental station with STM32

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1. Introduction

With the rapid growth of urbanization and industrialization, monitoring environmental indicators such as temperature, humidity, and gas concentration has become essential for public health and fire safety. Maintaining optimal indoor air quality is a critical necessity for preventing respiratory issues and ensuring early fire detection in residential and office spaces. Traditional monitoring systems are often bulky and expensive, making them impractical for small-scale applications. Embedded systems, particularly those powered by the STM32F103C8T6 microcontroller, offer a dedicated, low-power solution for real-time sensing and processing. By utilizing this technology, it is possible to create a compact device that handles high-speed sensor data without the complexity or cost of a general-purpose computer.

Manual environmental tracking is often inconsistent and fails to provide the immediate warnings necessary during critical events, such as gas leaks or extreme humidity fluctuations. Current consumer-grade devices frequently lack a cohesive balance between affordability and accessibility, often providing either only a local display or only a remote connection. This project addresses the need for a compact, low-cost yet highly reliable embedded system that bridges this gap. The core problem solved is the lack of a unified desktop station that offers dual-monitoring capability: local feedback for immediate checks and remote tracking for convenience. Furthermore, the system must overcome the limitations of passive monitoring by integrating active audible alerts for abnormal conditions.

The primary objective of this project is to develop a "Mini Environmental Monitoring Station" that serves as a comprehensive practice of embedded system design. The technical goals include integrating various sensors for real-time measurement, specifically the DHT11 for temperature and humidity and the MQ2 for gas and smoke detection. The station aims to implement local data visualization via an I2C LCD and remote tracking through a Bluetooth (HC-05) connection to a mobile application. Additionally, the project requires the design of an integrated buzzer alarm to provide immediate alerts for hazardous environmental conditions. Beyond software logic, the objective is to complete a full hardware development cycle, encompassing peripheral communication protocols such as I2C, UART, and ADC from schematic design to PCB fabrication.

The scope of this project is limited to an indoor desktop monitoring station designed for residential or office environments. While the system integrates multiple professional communication protocols, it is important to note certain technical limitations. The MQ2 sensor is a low-cost module and is not intended for industrial-grade ppm accuracy; it serves as a relative indicator for safety thresholds. Similarly, the DHT11 sensor provides humidity and temperature readings suitable for consumer use but has a lower sampling rate and accuracy range compared to laboratory-grade equipment. This project focuses on the implementation of a functional embedded prototype rather than a final commercial product.

This report is organized to follow the standard engineering design process. Chapter 2 defines the functional and non-functional requirements that the system must satisfy. Chapter 3 provides the detailed technical specifications for the hardware and software subsystems. Chapter 4 describes the system partitioning and the overall architecture, including the core block diagrams. Chapter 5 details the hardware design, covering the power tree, schematics, and PCB layout. Chapter 6 explains the firmware workflow and logic implementation. Chapter 7 presents the integration process and the results of the final system testing. Finally, Chapter 8 concludes the project and outlines potential future improvements.

In the final implementation, the gas density value shown in ppm is a scaled approximate value used for demonstration and threshold-based safety indication rather than an industrially calibrated gas concentration measurement.

2. System Requirements

2.1. Functional Requirements

1. Environmental sensing

FR-1.1 The system shall acquire environmental measurements.

FR-1.1.1 The system shall measure ambient temperature. Temperature shall be reported in °C.

FR-1.1.2 The system shall measure indoor relative humidity. Humidity shall be reported in %RH.

FR-1.1.3 The system shall measure indoor gas/smoke level. Gas/smoke level shall be reported as ppm.

2. User interaction and alerting

FR-2.1 The device shall provide user interaction through two push buttons.

FR-2.1.1 BUTTON_1 shall switch the gas alarm threshold between two preset values for demonstration: 660 ppm and 10 ppm.

FR-2.1.2 BUTTON_2 shall cycle the LCD display mode through four fixed screens.

FR-2.1.3 The four LCD modes shall be: (1) welcome screen, (2) gas density, (3) temperature, and (4) humidity.

3. Local display

FR-3.1 The system shall present measured information locally on a 16x2 LCD.

FR-3.1.1 Mode 1 shall display a welcome screen with line 1 “Envir Station” and line 2 “Group 8”.

FR-3.1.2 Mode 2 shall display gas density with label and unit in ppm.

FR-3.1.3 Mode 3 shall display temperature with label and unit in degree Celsius.

FR-3.1.4 Mode 4 shall display humidity with label and unit in percentage relative humidity.

4. Mobile data reporting

FR-4.1 The system shall transmit environmental information wirelessly through the HC-05 Bluetooth UART module.

FR-4.1.1 The transmitted data shall include temperature, humidity, and gas density.

FR-4.1.2 The firmware shall send three separate fixed-format messages: “Temp:XX.0*C”, “Humidity:XX.0%RH”, and “Gas density:XXX ppm”, followed by a separator line.

2.2. Non-Functional Requirements

1. Usability

NFR-1.1. The system shall be easy to use for non-technical users.

NFR-1.1.1. Simple UI: The system shall provide simple user interaction. The system shall use less than or equal to 2 buttons for user control (e.g., Display mode, Alarm acknowledge).

NFR-1.2. Readability: The system shall present values clearly.

NFR-1.2.1. Text on LCD shall be readable at 30-50 cm viewing distance.

NFR-1.3. The system shall meet basic electrical safety for a desktop device.

NFR-1.3.1. Electrical safety: The system shall not expose hazardous voltages to the user. User-accessible parts shall operate at less than or equal to 5 V DC (or be fully enclosed).

NFR-1.4. Thermal safety: The system shall not become dangerously hot in normal use.

NFR-1.4.1. Enclosure surface temperature shall be less than 50 degrees C during operation.

NFR-1.5. The system shall be low-cost.

NFR-1.5.1. Total hardware cost shall be less than or equal to 250,000 VND.

NFR-1.6. The system shall be compact for desktop use.

NFR-1.6.1. Size: The enclosure shall fit within 20 cm x 10 cm x 10 cm. Mass: Total mass shall be less than or equal to 300g.

2. Durability

NFR-2.1. The system shall operate reliably in indoor environments.

NFR-2.1.1. Temperature: The system shall operate from 0 to 40 degrees C (typical indoor). Humidity: The system shall operate at 20-80%RH non-condensing.

3. Reliability

NFR-3.1. The system shall run continuously with minimal failures.

NFR-3.1.1. Continuous operation: The system shall operate continuously for greater than or equal to 24 hours without manual reset.

NFR-3.2. Fault recovery: If a sensor read fails, the system shall continue running and show an error indicator.

NFR-3.2.1 Software robustness: If a sensor read fails, the firmware shall continue running, preserve the remaining system functions, and report the fault through the debug interface.

4. Performance

NFR-4.1. The system shall provide timely updates and alerts.

NFR-4.1.1. The system shall provide timely sensor update rate.

NFR-4.1.2. Temperature shall update at least once every 10 s. Humidity shall update at least once every 10-20 s. Gas/smoke level shall update at least once every 5 s.

NFR-4.2. Display update: LCD shall refresh at least once every 1 s or when the displayed value changes.

NFR-4.2.1. Alert latency: Alarm shall activate within less than or equal to 5 s after threshold violation is detected.

NFR-4.3. The system shall have low power consumption.

NFR-4.3.1. Average power: Average power consumption shall be less than or equal to .Peak current: Peak current shall be less than or equal to, to avoid brownout.

NFR-4.4. The system shall support the selected power source.

NFR-4.4.1. The system shall operate from an external power source 5 V supply.

2.3. Requirement validation

The requirement validation process ensures that the final report reflects the implemented prototype rather than an idealized design. Each requirement is checked against either simulation results, breadboard measurements, wireless terminal outputs, or LCD observations. The final prototype demonstrates successful integration of sensing, display, alert, and wireless transmission functions. In addition, the validation process confirms that the two-button user interface behaves consistently with the implemented firmware: BUTTON_1 changes the gas alarm threshold, and BUTTON_2 cycles the LCD display modes. Requirements that were not fully demonstrated in Proteus, such as perfectly stable DHT11 behavior or exact calibrated gas ppm, are discussed honestly as implementation limitations rather than overstated as fully validated specifications.

3. System Specifications

3.1. Product Specification

The Mini Environmental Station is a compact desktop embedded device for indoor environmental monitoring. The final prototype measures temperature and humidity using a DHT11 sensor and estimates gas concentration using an MQ2 analog sensor. Data is shown locally on a 16x2 I2C LCD and transmitted wirelessly through a HC-05 Bluetooth UART module to a mobile serial terminal application. The system also includes an active buzzer for gas-threshold warning and a heartbeat LED to indicate that the firmware is running normally. Two push buttons are used for user interaction: one changes the alarm threshold and the other cycles the LCD display mode.

3.2. Design Specification

The system operates as a periodic sensing-and-response loop. The STM32 acquires analog gas data from the MQ2 sensor through its 12-bit ADC and reads temperature/humidity values from the DHT11 using a timing-based single-wire routine. The firmware then updates the alarm state, refreshes the LCD according to the selected display mode, and transmits environmental values through the wireless UART interface. Unlike the earlier draft, the final implementation does not use a snooze function or Bluetooth enable/disable mode. Instead, BUTTON_1 changes the gas threshold between 660 ppm and 10 ppm for demonstration, while BUTTON_2 cycles the LCD through four fixed display screens.

3.3. Hardware Specification

Subsystem	Component	Main Function	Operating Voltage	Interface / Connection	Notes
Processing Unit	STM32F103C8T6 microcontroller	Central control, sensor reading, logic execution, display update, and wireless transmission	3.3 V	ADC, GPIO, I2C, UART, Timer	Main controller of the system

Temperature and Humidity Sensing	DHT11	Measures ambient temperature and relative humidity	3.3 V	1-wire digital interface via GPIO	Low-cost sensor for indoor monitoring
Gas Sensing	MQ-2 module	Detects gas/smoke level and provides analog output	5 V	Analog output to ADC input	Gas density value is approximate and used for demonstration and threshold warning
Local Display	16x2 LCD with PCF8574 I2C backpack	Displays system status and measured values locally	5 V	I2C bus	Reduces GPIO usage by using only SDA and SCL
Wireless Communication	HC-05 Bluetooth module	Sends environmental data wirelessly to smartphone or serial Bluetooth app	5 V	UART	Used for remote monitoring
Audible Alert	Active buzzer with transistor driver	Generates alarm sound when gas threshold is exceeded	5 V	GPIO output through transistor driver	Driver circuit protects the MCU from high current load
User Input	Push button 1	Changes gas alarm threshold between 660 ppm and 10 ppm	3.3 V logic	GPIO input	Used for demonstration of alarm behavior
User Input	Push button 2	Cycles the LCD display through 4 modes	3.3 V logic	GPIO input	Used for local interface control
Visual Status Indicator	LED	Heartbeat indicator showing the firmware is running	3.3 V logic	GPIO output	Toggles periodically during operation
5 V Regulation	LM2596 buck converter	Converts 9 V input to regulated 5 V rail	9 V input / 5 V output	Power stage	Supplies MQ-2, LCD, Bluetooth module, and buzzer
3.3 V Regulation	AMS1117-3.3 regulator	Converts 5 V rail to regulated 3.3 V rail	5 V input / 3.3 V output	Power stage	Supplies MCU and DHT11

Debug Interface	USB-to-TTL serial interface	Displays debug messages during development and testing	3.3 V logic UART	UART	Used during firmware testing and debugging
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The hardware specification of the final prototype includes the STM32 controller board, MQ2 analog gas sensor, DHT11 temperature-humidity sensor, 16x2 LCD with I2C backpack, active buzzer with transistor driver, HC-05 wireless UART module, two push buttons, and a two-stage power supply based on LM2596 and AMS1117. The hardware is designed to support ADC, I2C, UART, GPIO, and timer-based signal timing in a compact desktop monitoring platform.

3.4. Software Specification

The software is implemented using STM32 HAL and organized into an APP layer and a BSP layer. The BSP layer contains hardware-specific modules for LED, buzzer, buttons, gas sensing, DHT11, LCD, debug UART, and Bluetooth UART. The APP layer performs periodic sensing, threshold evaluation, LCD mode control, heartbeat LED operation, and wireless message formatting. This modular structure improves code readability, debugging, and subsystem reuse.

At startup, APP_Init configures the high-level system behavior by initializing the LCD, launching the timer used for DHT11 timing support, and preparing the status outputs. During runtime, APP_Run periodically updates the DHT11 reading, toggles the heartbeat LED, handles button events, refreshes the LCD according to the selected mode, updates the gas threshold alarm logic, and sends formatted environmental messages through the Bluetooth UART channel.

3.5. Test Specification

The system shall be tested in three stages: logic verification in Proteus, functional verification on breadboard hardware, and board-level validation on the fabricated PCB. Simulation is used mainly for digital logic and flow verification, while final confidence is obtained from real hardware tests, especially for LCD operation, UART communication, Bluetooth transmission, and threshold-based alarm response.

4. System partitioning and Architecture

4.1. System Partitioning

The Mini Environmental Monitoring Station is structured into five distinct functional subsystems to ensure high modularity and efficient firmware development. The Sensing Subsystem utilizes DHT11 and MQ2 sensors to capture ambient temperature, humidity, and gas data. These signals are directed to the Processing Subsystem, centered on the STM32F103C8T6 microcontroller, which is responsible for logic execution and threshold evaluation. Local interaction is managed by the User Interface Subsystem, featuring a 16x2 I2C LCD, an active buzzer for gas-threshold alerts, and buttons for threshold selection and display mode cycling. The Communication Subsystem utilizes a HC-05 Bluetooth UART module for wireless connectivity. Crucially, the firmware manages two separate UART channels: one dedicated to real-time debug messages and another for continuous environmental data transmission to a mobile application. Finally, the Power Subsystem provides a regulated energy backbone, converting the external 9V input into 5V and 3.3V rails to meet all hardware voltage requirements.

4.2. Overall block diagram

As illustrated in Figure 4.2, the overall system architecture is organized around the STM32F103C8T6 microcontroller, which serves as the central hub for data processing and subsystem coordination. The Power Supplier block manages the energy distribution, beginning with a 9V DC Adapter input that is regulated down to a 5V Rail via an LM2596 buck converter. This 5V supply is then further stepped down by an AMS1117 regulator to provide a stable 3.3V Rail for the MCU and sensors. This tiered power distribution ensures energy stability and provides a common ground for all functional modules.

The input layer integrates both environmental sensors and manual user controls. Specifically, the DHT-11 module provides digital humidity and temperature data via a GPIO interface, while the MQ2 module monitors gas and smoke levels using an Analog-to-Digital Converter (ADC). User interaction is facilitated through a dedicated button block, featuring a Threshold button and an LCD mode button—both connected via GPIO inputs to manage alarm threshold selection and display states.

For output and communication, the design incorporates a multi-interface strategy to provide comprehensive feedback. Local data visualization is achieved using an LCD Screen connected through an I2C bus to minimize pin usage. Remote reporting is established via a HC-05 module using a UART interface to transmit formatted environmental packets to a mobile application. Additionally, an audible alert system is implemented using a Buzzer driven by a dedicated Buzzer Driver circuit connected to a GPIO pin, ensuring immediate notification when environmental safety thresholds are exceeded.

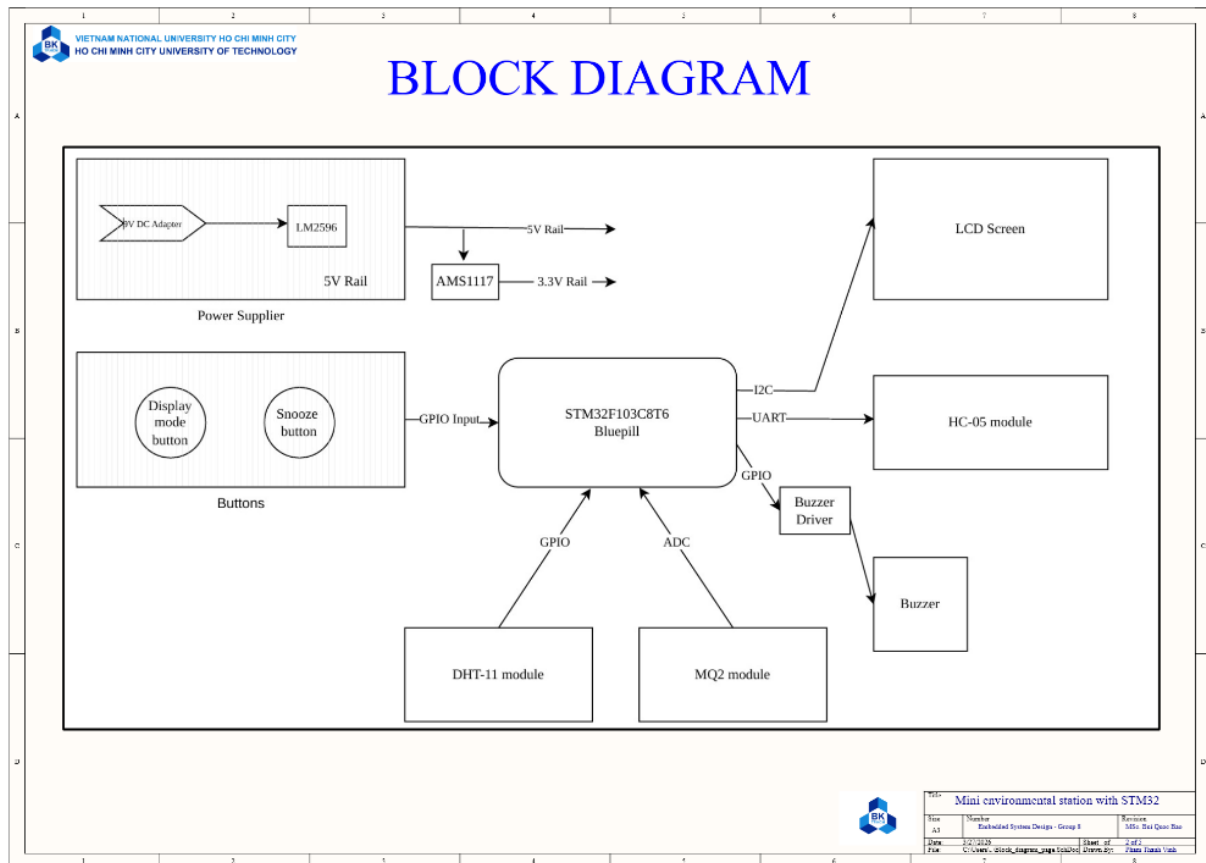


Figure 4.2.1: System Block Diagram

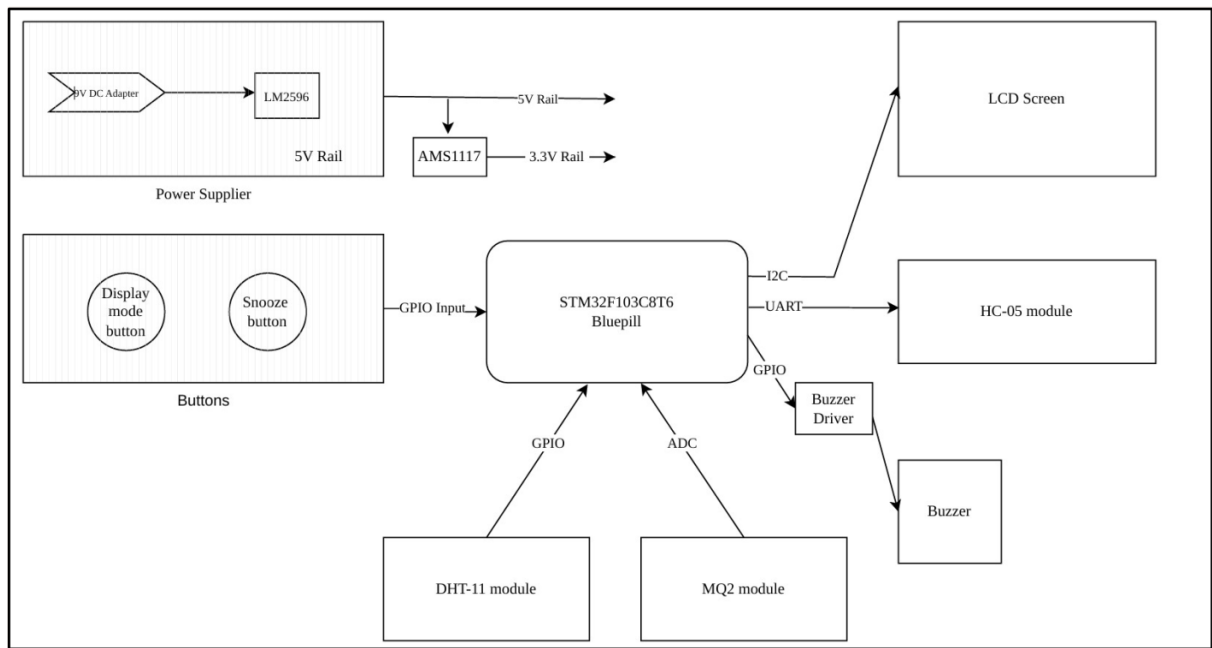


Figure 4.2.2: System Block Diagram

4.3. Data flow

The system data flow begins at the sensing layer, where the DHT11 provides temperature and humidity data and the MQ2 provides an analog gas-related voltage. The STM32 reads these inputs, converts them into software variables, and then executes three parallel output tasks: updating the LCD, controlling the buzzer according to the gas threshold, and transmitting environmental data through the HC-05 module. The control flow remains in the main firmware loop, while button actions are checked regularly to change the threshold or the LCD display mode. Unlike the earlier draft, the final implemented logic does not include a snooze alarm or Bluetooth transmission enable/disable mode.

4.4. Pin and interface allocation

MCU Pin / Peripheral	Peripheral	Interface Type	Description
PA0	MQ2 Sensor	Analog (ADC1_IN0)	Performs gas analog acquisition to estimate density.

MCU Pin / Peripheral	Peripheral	Interface Type	Description
PA1	DHT11 Sensor	Digital (GPIO)	Uses timer-based 1-wire routine for temperature and humidity reading.
PB6	LCD I2C	I2C1 (SCL)	Serial clock line for the 16x2 local display interface.
PB7	LCD I2C	I2C1 (SDA)	Serial data line for the 16x2 local display interface.
PB0	Buzzer Driver	Digital Output	Controls the active buzzer for threshold-based gas warnings.
PB12	Button 1	Digital Input	User interface button for gas alarm threshold selection (660/10 ppm).
PB13	Button 2	Digital Input	User interface button for cycling through the 4 LCD display modes.
UART (Debug)	Serial Terminal	UART	Dedicated channel for firmware development and real-time debugging.
UART (Wireless)	HC-05 Bluetooth	UART	Transmits separate messages for environmental values to mobile app.
TIM2	Internal Timer	Hardware Timer	Provides microsecond delay support for DHT11 signal timing.

5. Hardware Design

5.1. Design considerations

The design of the Mini Environmental Monitoring Station balances technical performance, affordability, and practical implementation for an academic embedded system project. To ensure the project objectives are met within the specified constraints, the selection of hardware components is guided by several critical criteria. The primary consideration is the availability of adequate hardware interfaces, specifically the requirement for integrated I2C, UART, ADC, and multiple GPIO channels to support simultaneous communication with various sensors and peripherals. Furthermore, the components must be easily accessible in the local market and cost-effective, adhering to a total budget limit of 250,000 VND. The hardware is also selected to be appropriate for the junior-level academic scope, offering a professional-grade platform that is nonetheless easy to simulate in software like Proteus and straightforward to prototype on a breadboard.

Beyond component selection, the hardware architecture incorporates fundamental embedded system design factors including cost, power, size, and reliability. Cost management is achieved by utilizing standardized modules such as the DHT11 and MQ2, which provide reliable environmental data at a low entry price. For power efficiency, the system utilizes an LM2596 buck converter to regulate the 9V adapter input to a stable 5V rail, minimizing thermal dissipation compared to standard linear regulators. The design is intentionally compact to facilitate desktop use, ensuring the final assembled device fits within the specified volume of 20 cm x 10 cm x 10 cm. To ensure operational reliability, the hardware is designed to support continuous 24-hour operation and includes considerations for software-based fault recovery mechanisms, such as a watchdog timer, to prevent system failure during long-term monitoring. These considerations ensure that the current design provides a robust and realistic solution for indoor air quality monitoring.

The next section will detail the specific selection and justification of the STM32F103C8T6 Microcontroller as the central processing unit for this architecture.

5.2. Microcontroller selection

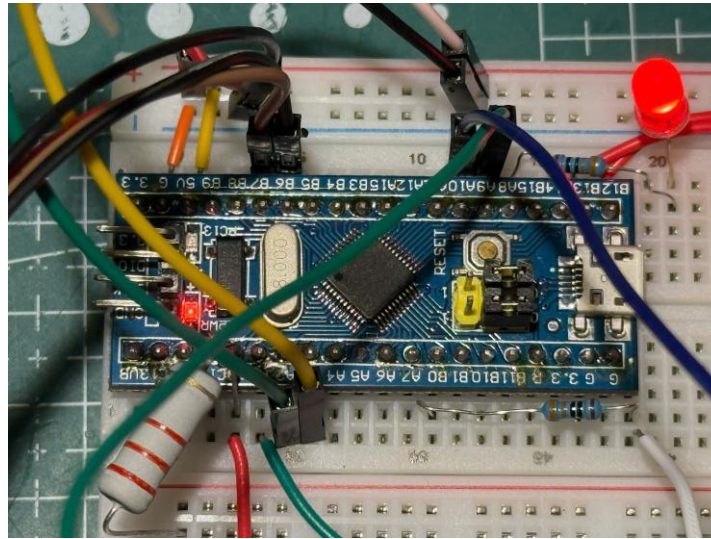


Figure 5.2: Microcontroller STM32F103C8T6

The STM32F103C8T6 microcontroller serves as the central processing unit and architectural hub for the Mini Environmental Monitoring Station. Tasked with executing the system's core logic, this 32-bit MCU manages real-time data acquisition from sensors, evaluates safety thresholds, and coordinates the output peripherals to ensure reliable user feedback. The selection of this specific platform was driven by its ARM Cortex-M3 architecture, which provides the necessary computational headroom for concurrent tasks—such as high-speed ADC sampling for the MQ2 sensor and I2C communication for the LCD—without performance bottlenecks. This processing capability directly supports the hardware configuration previously detailed, enabling the simultaneous management of the dual UART channels for debugging and HC-05 wireless transmission while maintaining stable timing for the DHT11 sensor.

Beyond raw performance, the microcontroller's rich set of integrated peripherals aligns perfectly with the project's hardware requirements. The availability of dedicated I2C, UART, and 12-bit ADC channels allows for a streamlined hardware design, minimizing the need for external interface components and reducing the overall PCB footprint. This high level of integration is essential for maintaining the device's compact desktop form factor while ensuring precision in environmental monitoring. Furthermore, the choice is highly cost-effective, directly supporting the project's strict budget constraint of less than 250,000 VND. Its suitability for the academic scope of the project, combined with its compatibility with simulation tools and breadboard-friendly layout, made it the ideal choice for iterative development and verification.

The resource allocation for this design utilizes the MCU's capabilities efficiently, ensuring zero-conflict operation between various functional subsystems. Specifically, pin PA0 is assigned to the MQ2 sensor's analog input, while PA1 handles the digital 1-wire protocol for the DHT11 sensor. Communication with the 16x2 LCD is managed via pins PB6 and PB7 using the I2C1 bus. The HC-05 Bluetooth module interfaces through a dedicated UART channel, allowing the system to transmit formatted environmental messages to the mobile application while maintaining a separate UART channel for real-time debugging. Additionally, the MCU's External Interrupt (EXTI) capabilities are leveraged on pins PB12 and PB13 to provide instantaneous responsiveness for the display mode. This configuration ensures that user inputs are processed with high priority, allowing for seamless interaction even during intensive sensor sampling cycles.

5.3. Power supply design

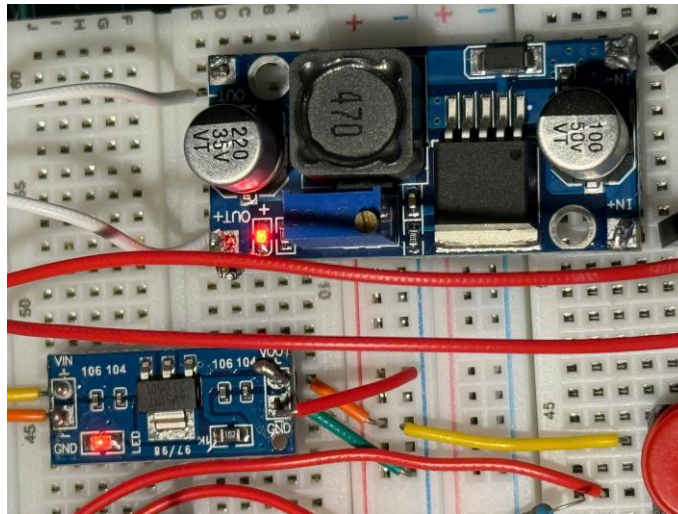


Figure 5.3.1: Power supply modules LM2596 and AMS1117

The power system is designed as a tiered distribution architecture to ensure voltage stability and energy efficiency for the entire monitoring station, as detailed in the Power Distribution Tree (Figure 5.1) and the Power Supply Schematic (Figure 5.2). The primary power source originates from a 9V DC Adapter (2A, 18W maximum power), which is subsequently regulated through two distinct step-down stages to meet the specific voltage requirements of each component. The first regulation stage utilizes an LM2596 buck converter to transform the 9V input into a 5V Rail.

As shown in the schematic in Figure 5.2 , this block features a 100uH inductor, an MSS1P4 diode, and a 330uF filtering capacitor system to optimize efficiency and minimize ripple voltage. The 5V Rail serves as the primary backbone, supplying power to high-power modules. Specifically, the power tree in Figure 5.1 shows that this rail powers the MQ-2 gas sensor (250mA peak), the HC-05 Bluetooth module (80mA peak), the 16x2 LCD with I2C interface (90mA peak), and the active buzzer (100mA peak). To provide a stable and low-noise supply for the logic processing unit, a second regulation stage is implemented using an AMS1117 linear regulator to convert 5V to a 3.3V Rail.

The schematic in Figure 5.2 illustrates that the 3.3V output is equipped with parallel filtering capacitors (4.7uF, 22uF, 1uF, and 100uF) to ensure signal purity. This rail is dedicated to the STM32F103C8T6 Bluepill microcontroller (35mA average) and the DHT11 sensor (~1mA average). This tiered distribution effectively isolates electrical noise generated by power-hungry modules like the MQ-2 and Buzzer, preventing voltage brownouts in the central processing block. The design's reliability is confirmed through the Thermal Check tables integrated into Figure 5.1. For the LM2596, at an average load of 0.30A, the junction temperature (T_j) is calculated to be only 26-33°C (with an ambient temperature $T_a = 25^\circ\text{C}$), which is well within the ideal operating range. Even when operating at a peak current of 0.60A, the T_j remains stable at 46-55°C. For the AMS1117, at a rated current of 80mA, the T_j reaches approximately 32-39°C. These parameters demonstrate that the system not only operates efficiently but also ensures absolute thermal safety, allowing the device to function continuously for at least 24 hours without the risk of overheating or voltage instability.

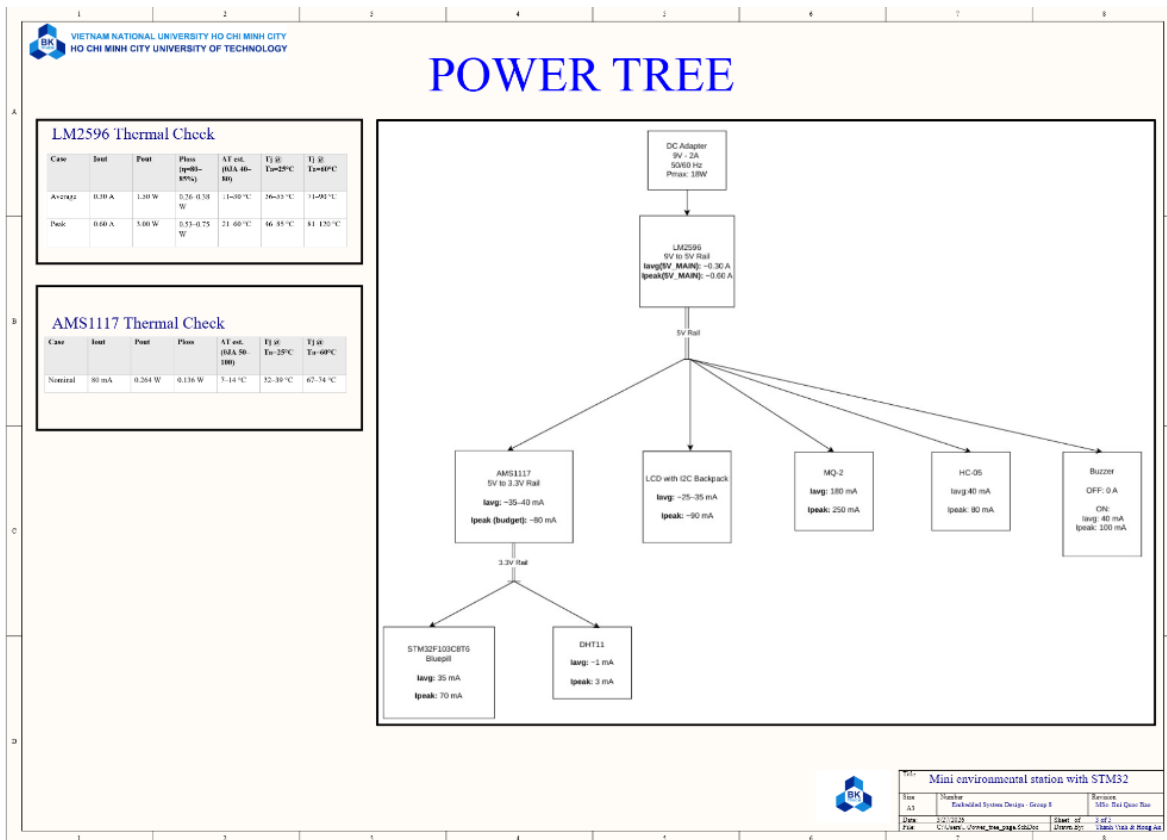


Figure 5.3.2: Power Distribution Tree (Full schematic page)

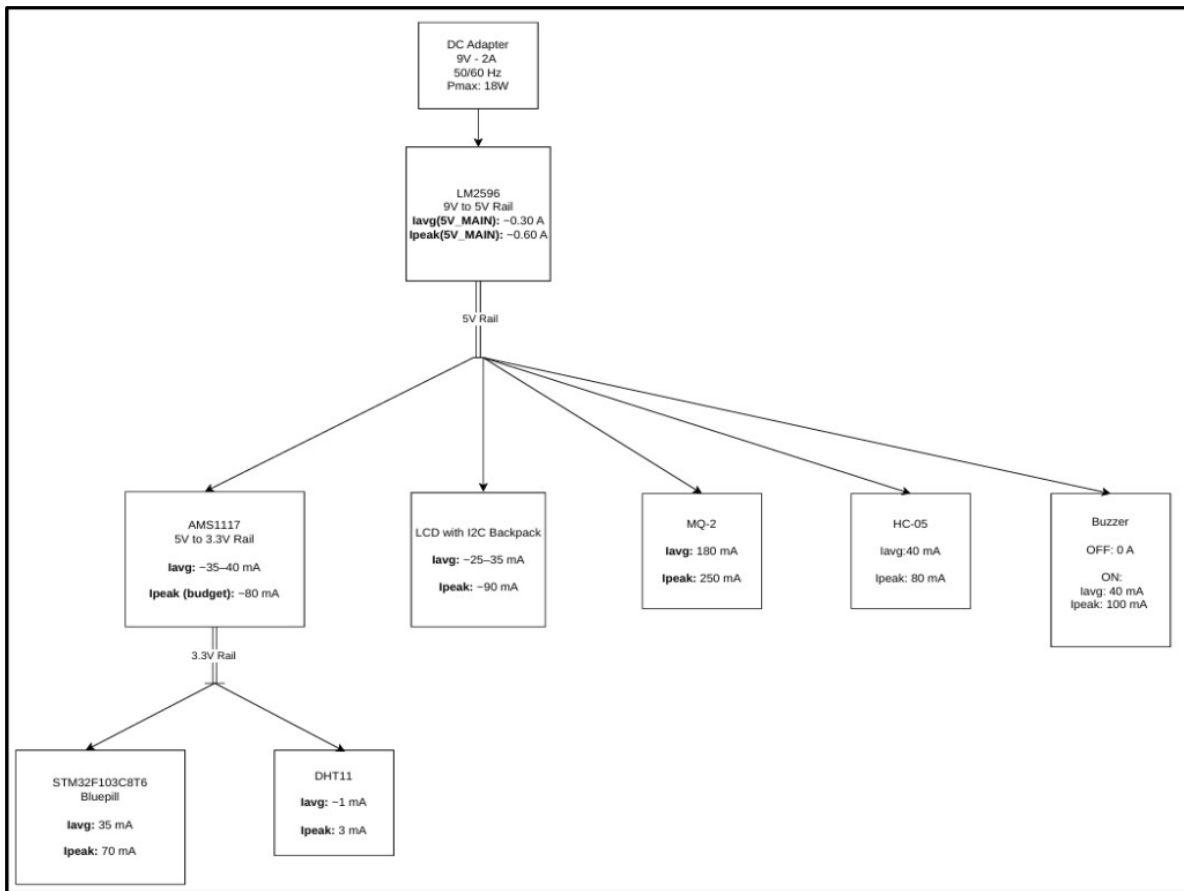
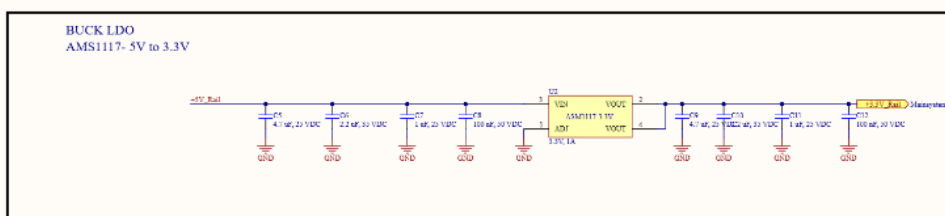
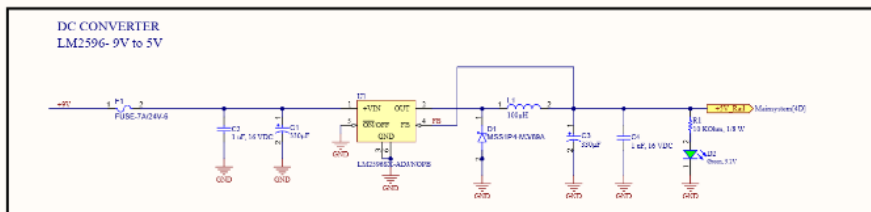
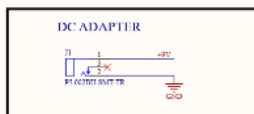


Figure 5.3.3: Power Distribution Tree (Zoom in)

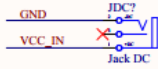
POWER SUPPLY



Mini environmental station with STM32			
File	Number	Revision	Author
AD	Embedded System Design - Group 5	1.0.0	Minh Thai Quan Rao
Drawn	12/11/20	Sheet of	4 of 1
File	C:\Users\MinhThai\Documents\Projects\Mini environmental station with STM32	Sheet of	4 of 1

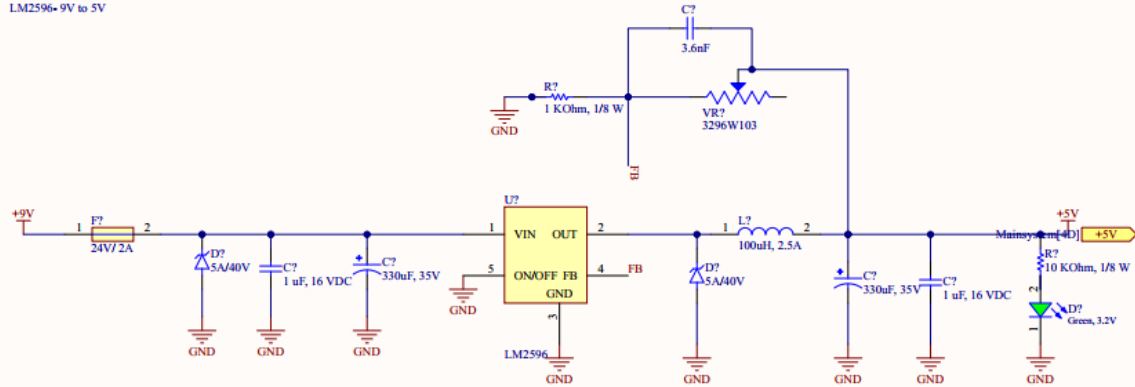
Figure 5.3.4 : Power Supply Schematic (Full schematic page)

DC ADAPTER



DC CONVERTER

LM2596- 9V to 5V



BUCK LDO

AMS1117- 5V to 3.3V

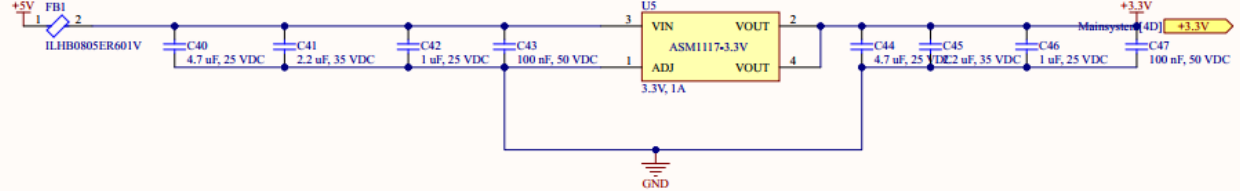


Figure 5.3.4 : Power Supply Schematic (Zoom in)

5.4. Sensor interface design

The sensor interface design focuses on establishing reliable data acquisition from the physical environment using both digital and analog protocols. This layer ensures that ambient variables are accurately captured and converted into a format suitable for the STM32 processing core.

5.4.1. DHT11 interface

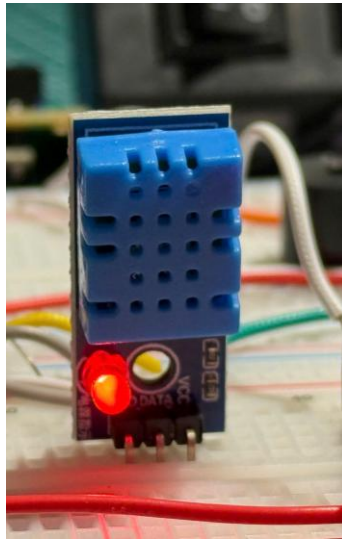


Figure 5.4.1 : DHT-11 Sensor

The DHT11 sensor is utilized to monitor indoor temperature and relative humidity. It employs a proprietary 1-wire digital protocol, where a 40-bit data packet is transmitted to the MCU through a single GPIO pin, PA1. The interface requires precise timing to distinguish between high and low bits representing the environmental data. To ensure signal stability, the data line is typically pulled high via a resistor, allowing the sensor to drive the line low to initiate communication. This configuration provides humidity readings in %RH and temperature in degrees Celsius (°C).

5.4.2. MQ2 analog interface



Figure 5.4.2 : MQ-2 Sensor

The MQ2 module is integrated to detect the presence of combustible gases and smoke. Unlike the digital DHT11, the MQ2 provides a continuous analog voltage level proportional to the gas concentration in the air. This signal is routed to the STM32 pin PA0, which is configured as an Analog-to-Digital Converter (ADC1_IN0) input. The MCU utilizes its internal 12-bit ADC to sample this voltage, subsequently applying mathematical mapping to determine the gas density in parts per million (ppm). This interface is critical for the safety subsystem, as it provides the raw data needed to evaluate against the 800 ppm alarm threshold.

5.5. User interface and communication circuits

5.5.1. LCD I2C interface



Figure 5.5.1 : LCD Screen with I2C backpack PCF8574

Local data visualization is achieved through a 16x2 LCD screen. To minimize the number of required GPIO pins, the display is connected via an I2C backpack based on the PCF8574 remote 8-bit I/O expander. The interface uses only two serial lines: SCL connected to PB6 and SDA connected to PB7. This configuration allows the STM32 to transmit formatted character strings, such as "Temp:XX.X*C" and "Humidity:XX.X%RH", to the screen for immediate display.

5.5.2. Button interface

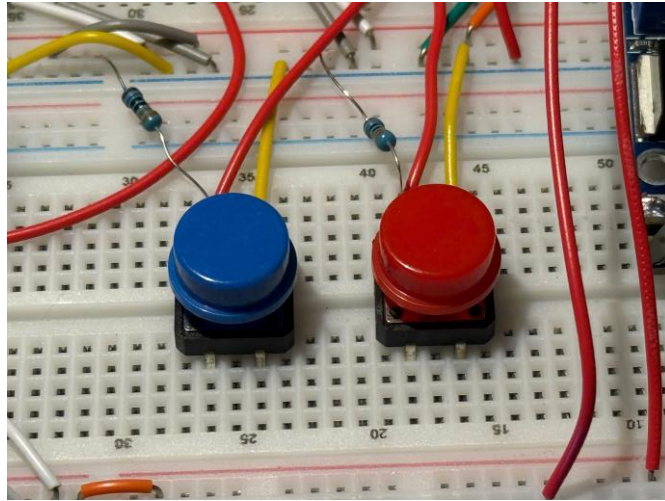


Figure 5.5.2 : Buttons and their drivers

The system uses two push buttons. BUTTON_1 is connected to PB12 and is used to change the gas alarm threshold between 660 ppm and 10 ppm. BUTTON_2 is connected to PB13 and is used to cycle the LCD through four display modes. This design provides a simple and effective demonstration interface without increasing hardware complexity.

5.5.3. Buzzer driver

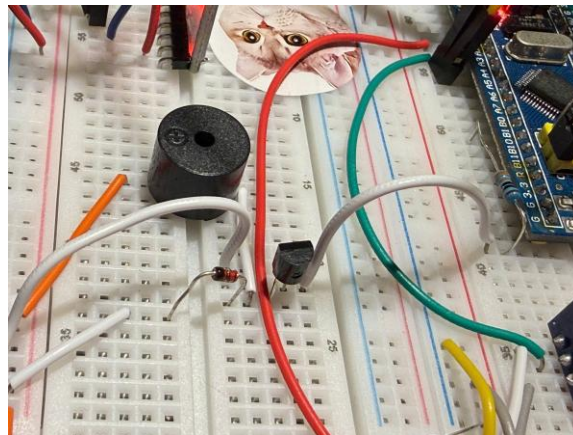


Figure 5.5.3 : Buzzer and its drivers

The audible alert system uses an active buzzer to notify users of hazardous conditions. Since the buzzer requires more current than a standard MCU GPIO can safely provide, a dedicated buzzer driver circuit is implemented. This circuit uses a transistor to act as an electronic switch, controlled by pin PB0. When the firmware detects a threshold violation (> 660 ppm), it sets PB0 high to saturate the transistor and activate the alarm.

5.5.4. HC-05 UART interface

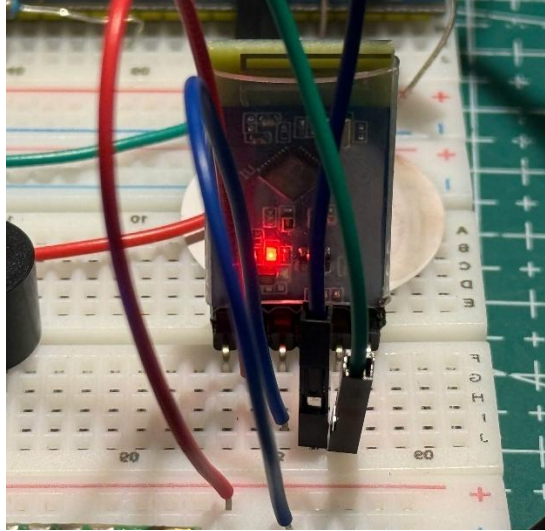


Figure 5.5.4 : Bluetooth module

Remote monitoring is established via an HC-05 Bluetooth module using a UART interface. The module's TX and RX pins are cross-connected to the MCU's PA3 (RX) and PA2 (TX) pins respectively. The system transmits formatted data frames containing temperature, humidity, and gas density values at a standard baud rate to a paired mobile application. This interface enables the "Mini Environmental Monitoring Station" to serve as a IoT-ready device for remote surveillance.

5.6. Complete schematic design

The overall system schematic is presented in Figure 5.3, illustrating all electrical and logic connections between the central microcontroller and the peripheral modules. The design is organized into distinct functional blocks to ensure transparency, providing optimal support for the system testing and maintenance processes.

At the heart of the circuit is the STM32F103C8T6 microcontroller, equipped with essential support circuits for stable operation. This block includes external crystal oscillators (HSE 8MHz and LSE 32.768kHz), a hardware reset circuit integrated with pushbutton SW1, and decoupling capacitors (C18, C19, C20) positioned near the MCU power pins. The BOOT0 pin is configured via a resistor to determine the system's boot mode.

The sensor and input block is designed to collect environmental data accurately. The DHT11 sensor connects to pin PA1 through a pull-up resistor R12 to maintain signal integrity for digital 1-wire communication. The MQ2 sensor connects to analog pin PA0 using a voltage divider circuit with load resistor R10 to convert gas concentration into a voltage signal compatible with the MCU's 12-bit ADC. Additionally, two interactive buttons are equipped with pull-up resistors R8 and R9: BUTTON 1 is dedicated to gas alarm threshold selection, and BUTTON 2 is used to cycle through LCD display modes.

Display and communication interfaces are integrated through standard buses to optimize hardware resources. The LCD screen uses an I2C converter module (PCF8574) connected to pins PB6 and PB7, accompanied by pull-up resistors R2 and R3 to ensure data transmission stability. The wireless communication block employs a HC-05 Bluetooth module connected via a dedicated UART channel, enabling the continuous transmission of formatted environmental data frames to the mobile application.

The audible alarm system utilizes a buzzer controlled by an NPN transistor (Q1) through current-limiting resistor R6. This circuit is protected by diode D3 to suppress flyback pulses generated during buzzer switching. Finally, the design integrates a status LED (DS1) and a UART_DEBUG header, which was retained for development and verification, facilitating real-time monitoring and direct hardware debugging during the system integration phase.

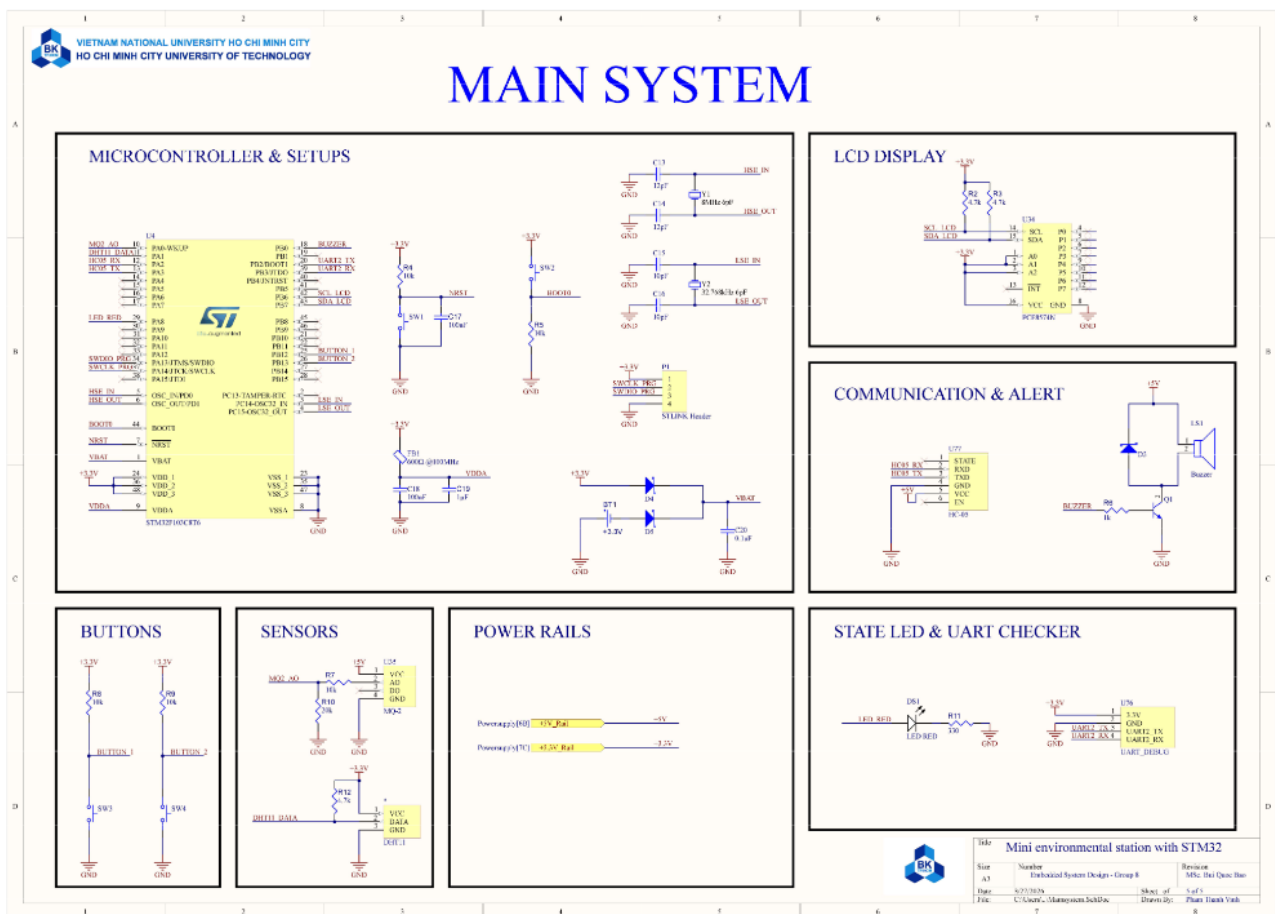


Figure 5.6.1: Overall System Schematic (Full schematic page)

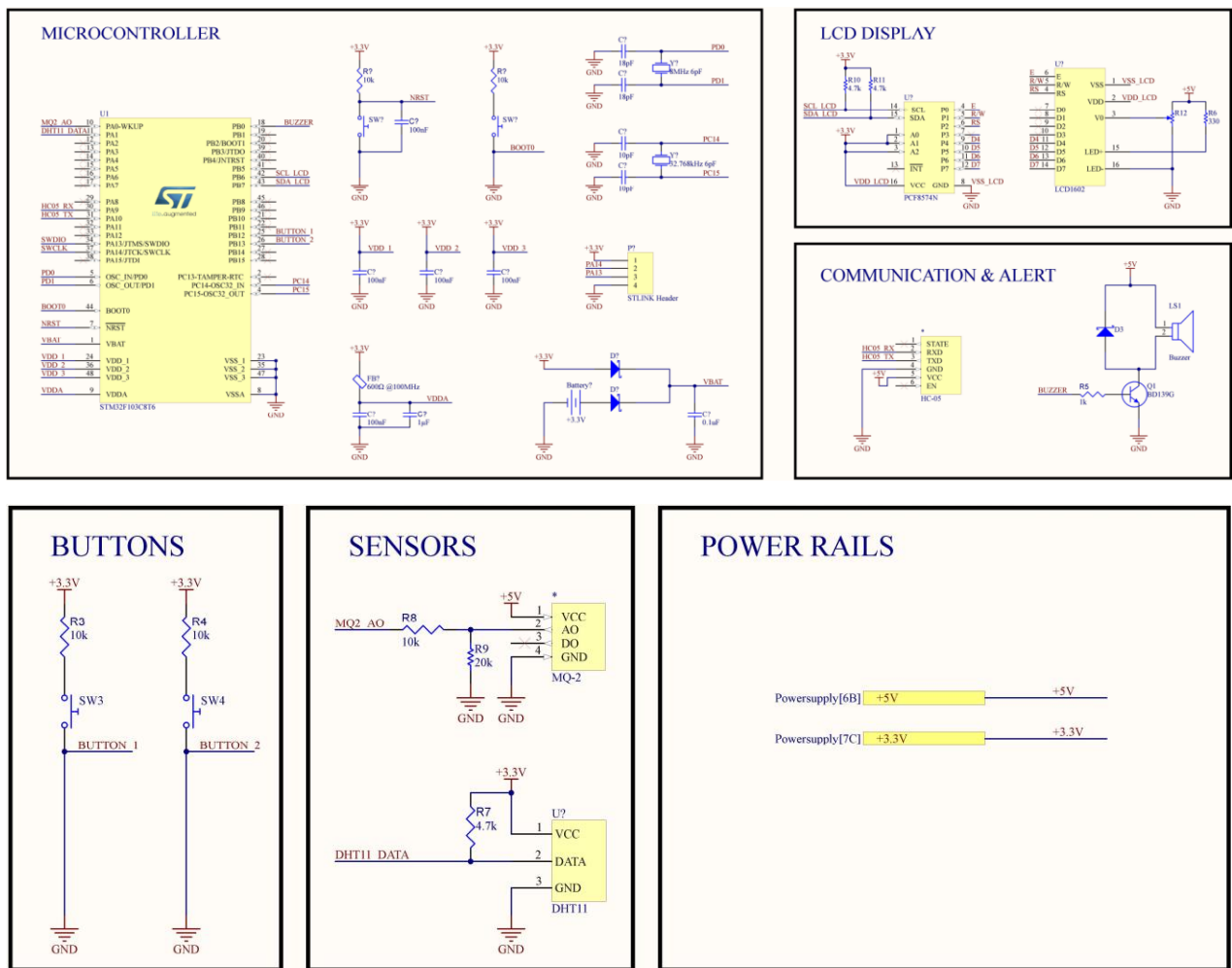


Figure 5.6.2: Overall System Schematic (Zoom in)

5.7. PCB layout design

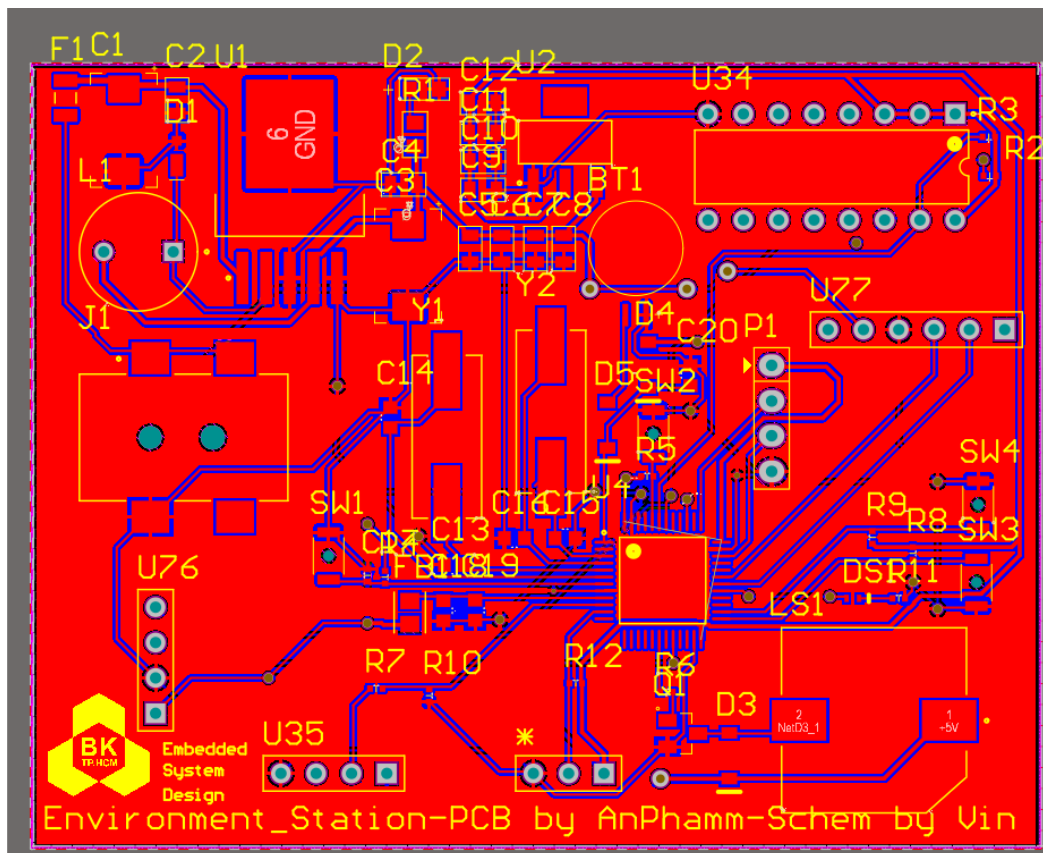


Figure 5.7.1: 2D PCB Layout

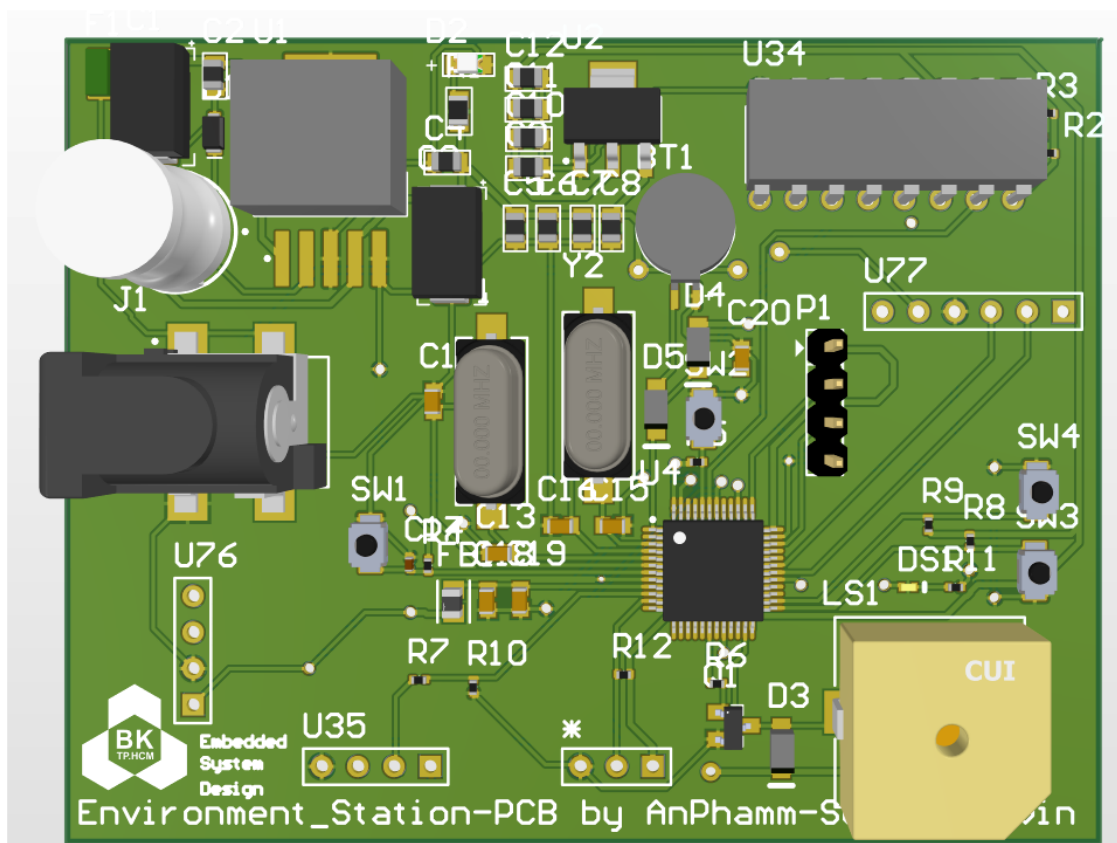


Figure 5.7.2: 3D PCB Render

5.8. Prototype and fabricated hardware

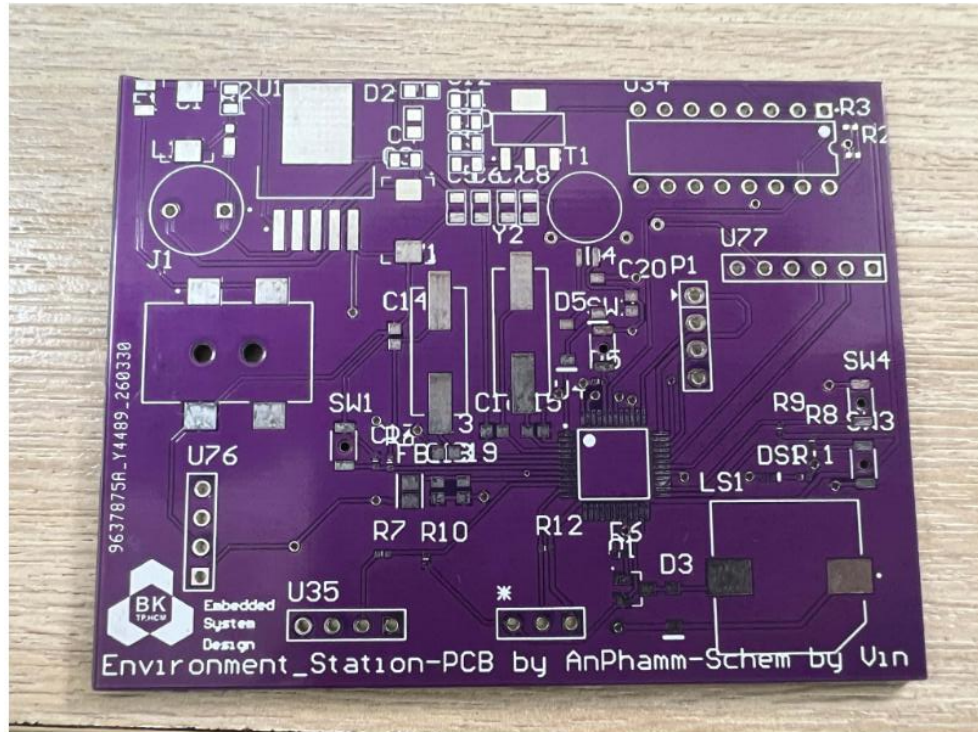


Figure 5.8.1: Fabricated PCB - Top Side

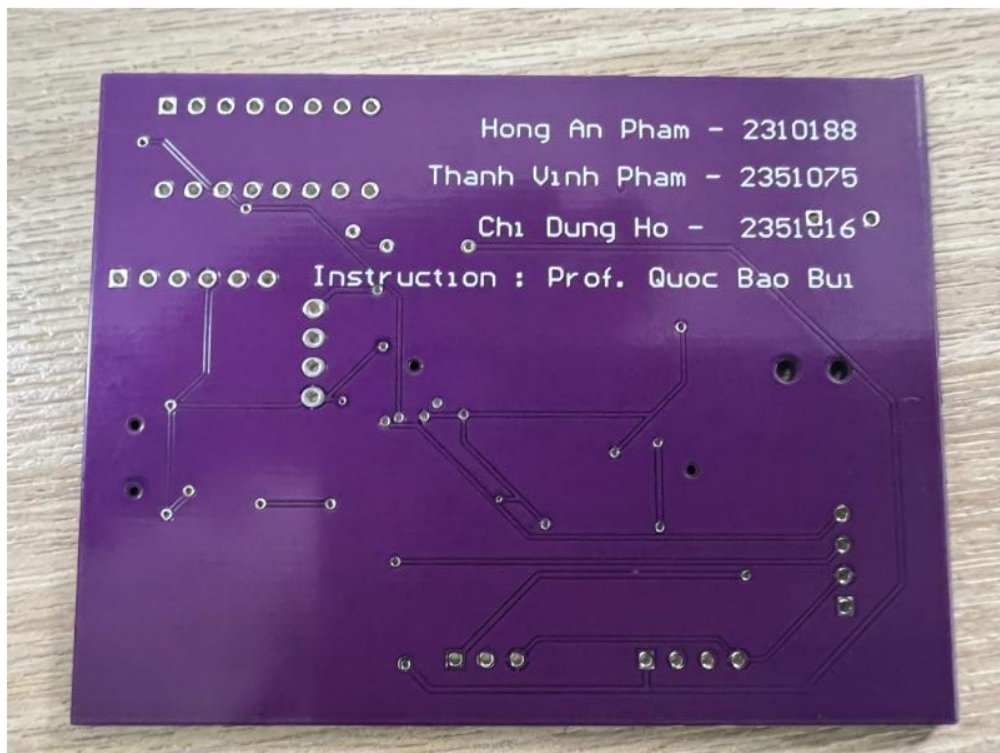


Figure 5.8.2: Fabricated PCB - Bottom Side

6. Firmware Design

6.1. Firmware architecture

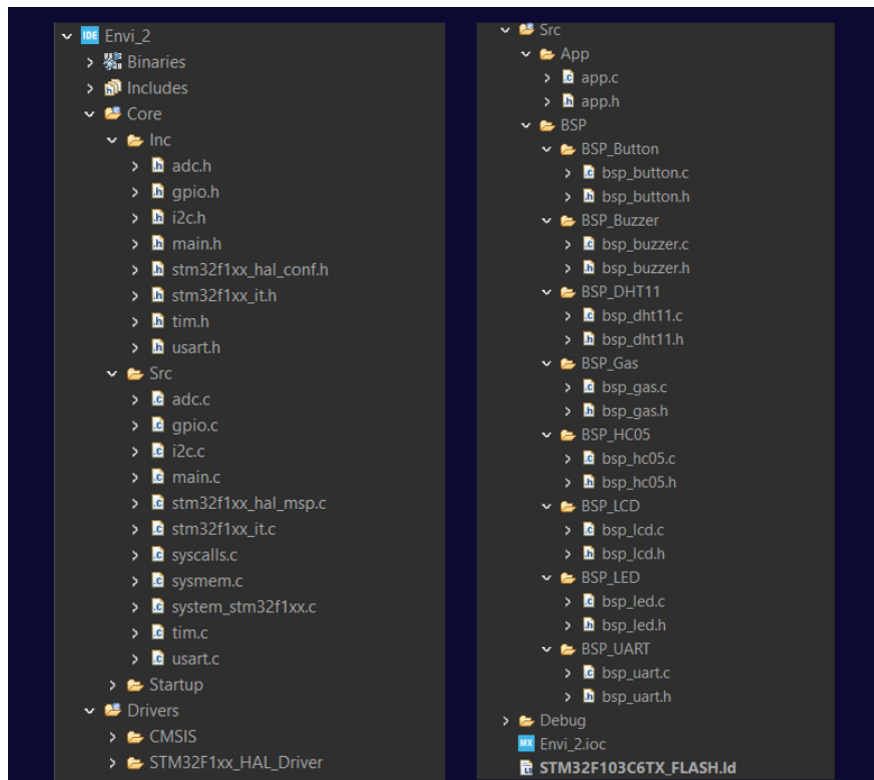


Figure 6.1: Firmware structure

The final firmware is structured into two layers: a Board Support Package (BSP) layer and an Application (APP) layer. The BSP layer contains low-level functions for each hardware block, such as LED, buzzer, buttons, gas sensor, LCD, DHT11, debug UART, and wireless UART. The APP layer combines these modules into the final system behavior, including threshold switching, display mode management, heartbeat signaling, sensor reading, alarm control, and wireless transmission. This organization reduces coupling between hardware details and system logic, making the code easier to maintain and extend.

6.2. Peripheral configuration

The firmware configures the ADC for MQ2 gas acquisition, I2C for LCD communication, timer-based microsecond delay support for DHT11 communication, one UART for debug messages, and one UART for the HC-05 wireless module. GPIO lines are used for the buzzer driver, two push buttons, the heartbeat LED, and the DHT11 data line. The code is generated

and managed in STM32CubeIDE using the HAL framework, with application-specific logic implemented on top of the generated initialization code.

6.3. Main firmware workflow

After initialization, the firmware enters an infinite loop. The loop periodically attempts to read the DHT11 sensor, toggles the heartbeat LED, checks button events, updates the gas alarm logic, refreshes the LCD display, and sends environmental messages through the wireless UART. `BUTTON_1` changes the gas threshold between 660 ppm and 10 ppm, while `BUTTON_2` cycles the LCD mode. The gas sensing routine continuously updates the approximate gas density value, compares it against the current threshold, and activates or deactivates the buzzer accordingly.

6.4. Sensor acquisition logic

The sensor acquisition layer serves as the primary data source for the system, responsible for converting physical environmental changes into digital values that the STM32F103C8T6 can process. This logic is partitioned into two distinct routines based on the specific communication protocols required by the integrated hardware components. The system architecture ensures a closed-loop process where raw signals from the sensing layer are captured and mathematically converted into standardized units of measure, such as degrees Celsius, percentage relative humidity, and parts per million.

6.4.1. DHT11 read sequence

The acquisition logic for the DHT11 follows a strict timing sequence required by its proprietary 1-wire protocol. To initiate communication, the MCU sends a start signal by pulling the PA1 pin low for at least 18ms and then releasing it to high. Following this, the MCU switches the pin to input mode and waits for the DHT11 to provide a response signal, indicating it is ready to transmit data. The sensor then transmits a 40-bit packet consisting of 16 bits for humidity, 16 bits for temperature, and an 8-bit checksum used for data integrity verification. If the timing between received bits exceeds the expected duration or if the checksum does not match the data, the firmware implements fail-handling procedures by discarding the reading and maintaining the last valid value.

In simulation, the DHT11 model showed some instability, so the final confidence in the implementation was obtained through real hardware behavior and code integration rather than simulation alone.

6.4.2. MQ2 ADC sampling

The MQ2 sensor logic relies on the STM32's internal Analog-to-Digital Converter (ADC) to interpret ambient gas and smoke concentrations. The firmware initiates a conversion on the ADC1_IN0 channel connected to pin PA0, allowing the MCU to capture a 12-bit raw digital value representing the analog voltage levels from the sensor. This raw value is compared against a calibrated digital threshold that corresponds to the 800 ppm safety limit. The MQ2 sensor is connected to the MCU through an ADC channel. The firmware reads the analog voltage, performs software scaling, and generates an approximate gas density value for demonstration and threshold-based warning. This value is sufficient for comparing relative gas levels and for triggering the alarm logic in the final prototype.

6.5. Alarm and display logic

The final prototype activates the buzzer when the gas density value exceeds the currently selected threshold. BUTTON_1 allows the demonstrator to switch between a high threshold of 660 ppm and a very low threshold of 10 ppm, making it easy to show both normal and alarm states during presentation. BUTTON_2 cycles the LCD through four modes: a welcome screen, gas density, temperature, and humidity. In addition, a heartbeat LED toggles periodically to show that the firmware is alive even when no button is pressed.

6.6. Bluetooth transmission logic

The wireless subsystem sends environmental information through the HC-05 module. Instead of packaging all values into one long sentence, the final firmware transmits three separate fixed-format lines: one line for temperature, one line for humidity, and one line for gas density. A separator line is transmitted after each update group to improve readability on the mobile serial Bluetooth terminal.

6.7. Error handling and robustness

The firmware is designed so that a sensor read failure does not stop the whole system. If the DHT11 read routine fails, an error message is printed to the debug UART, while the remaining functions—LCD mode handling, buzzer control, gas sensing, and wireless transmission—continue to operate. This simple fault containment strategy increases the robustness of the final prototype without adding excessive software complexity.

7. Integration and Testing

7.1. Integration strategy

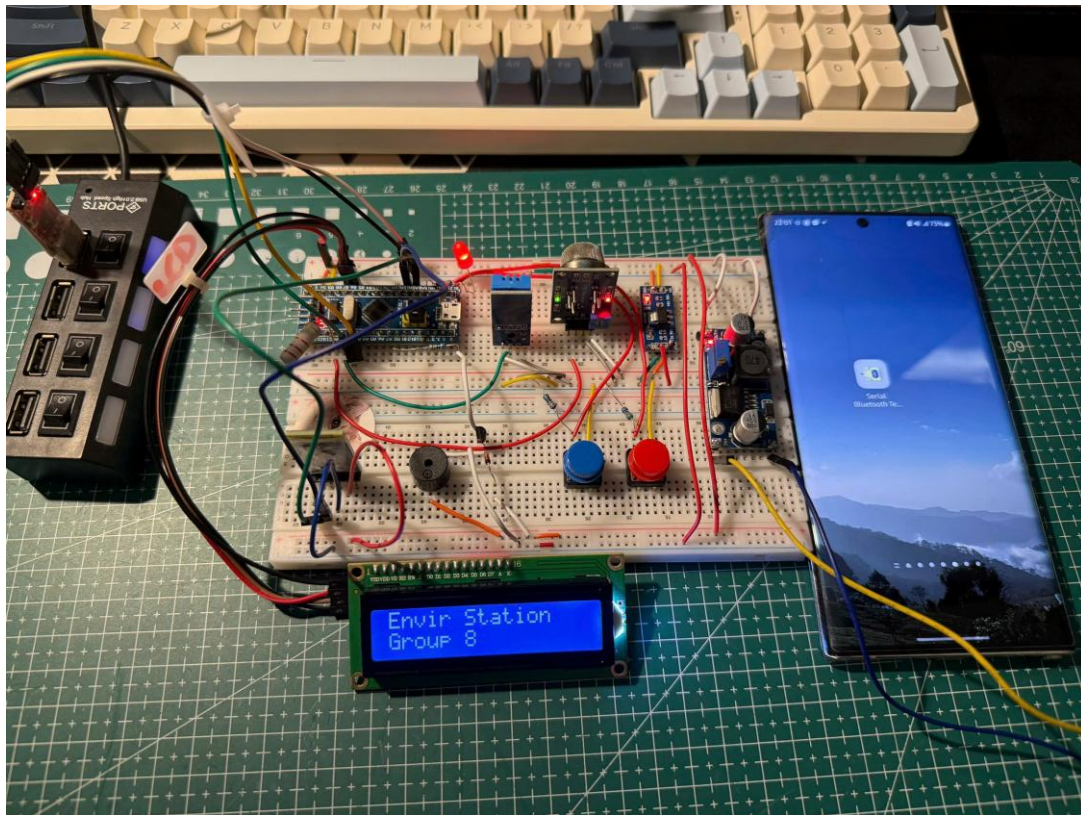


Figure 7.1: Full system integration on breadboard

The integration process was carried out incrementally. The team first verified basic GPIO and UART communication, then validated the buzzer and button behavior, followed by ADC-based gas sensing. LCD I2C communication was then confirmed on real breadboard hardware. Finally, the wireless transmission channel and the complete APP–BSP integrated firmware were validated. This staged approach reduced debugging complexity and made it easier to isolate subsystem errors before the final full-system test.

7.2. Simulation results

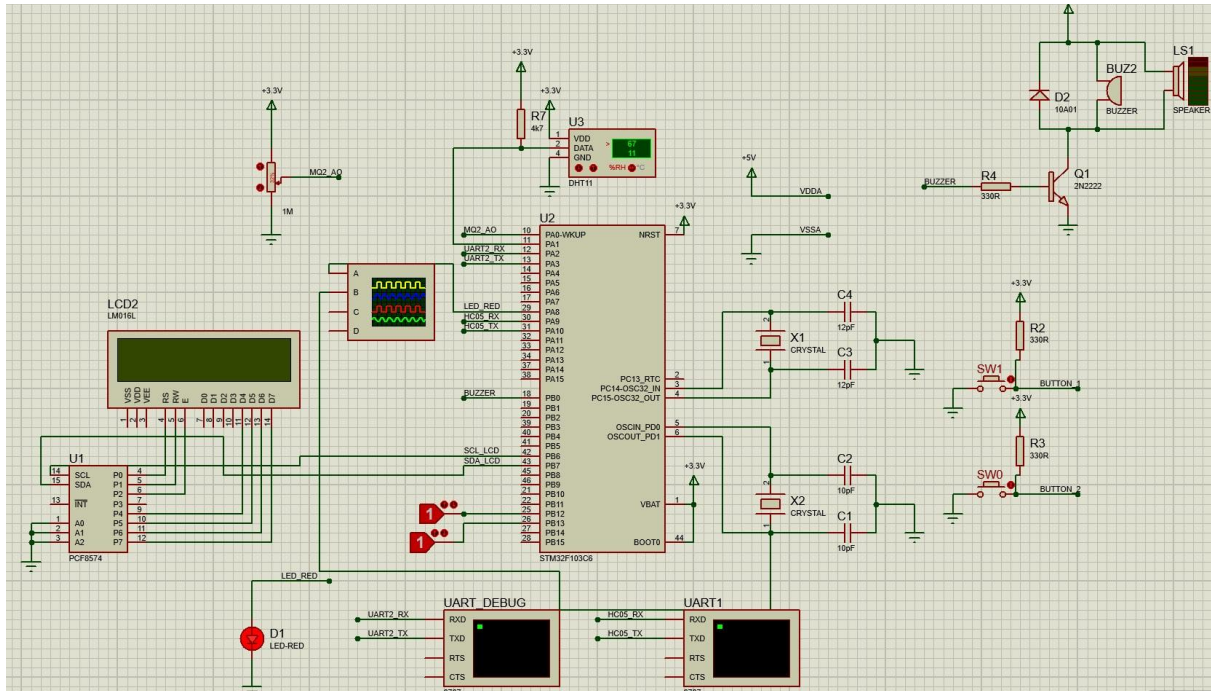


Figure 7.2.1: The simulation module in Proteus

Proteus simulation was primarily used to verify logic flow, GPIO control, ADC scaling behavior, buzzer control, and UART message formatting. Some simulated peripheral models, especially DHT11 and LCD I2C, were less reliable than the corresponding real hardware. Therefore, simulation was used mainly as an intermediate verification step rather than the sole source of final validation.

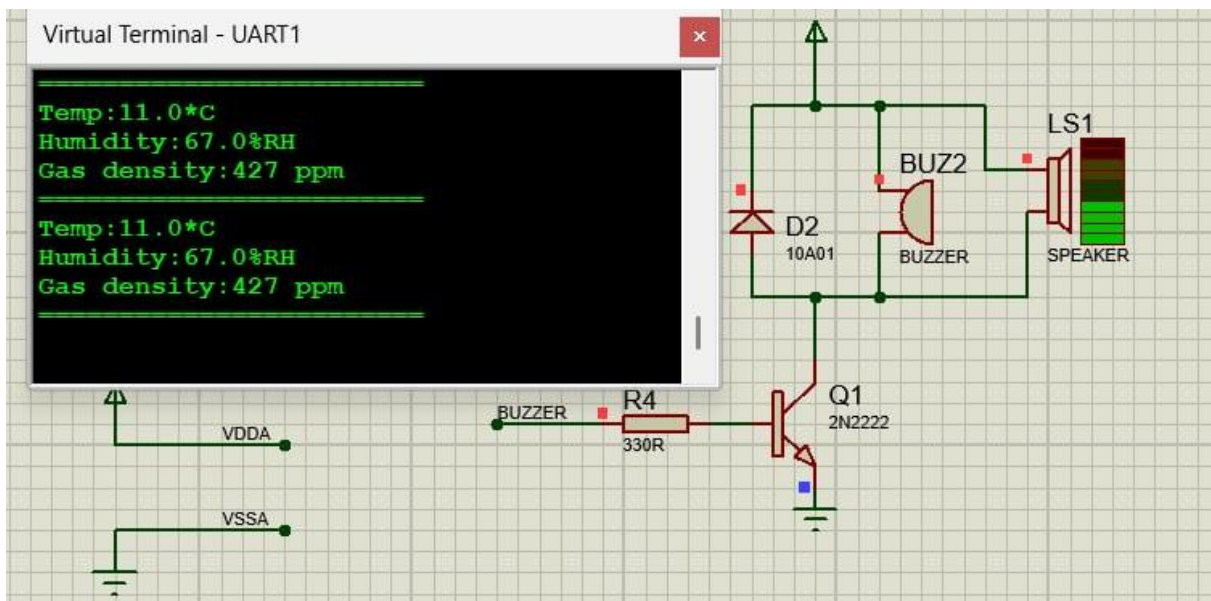


Figure 7.2.2: Successful simulation result with buzzer on and HC-05 sending messages

7.3. Breadboard test results

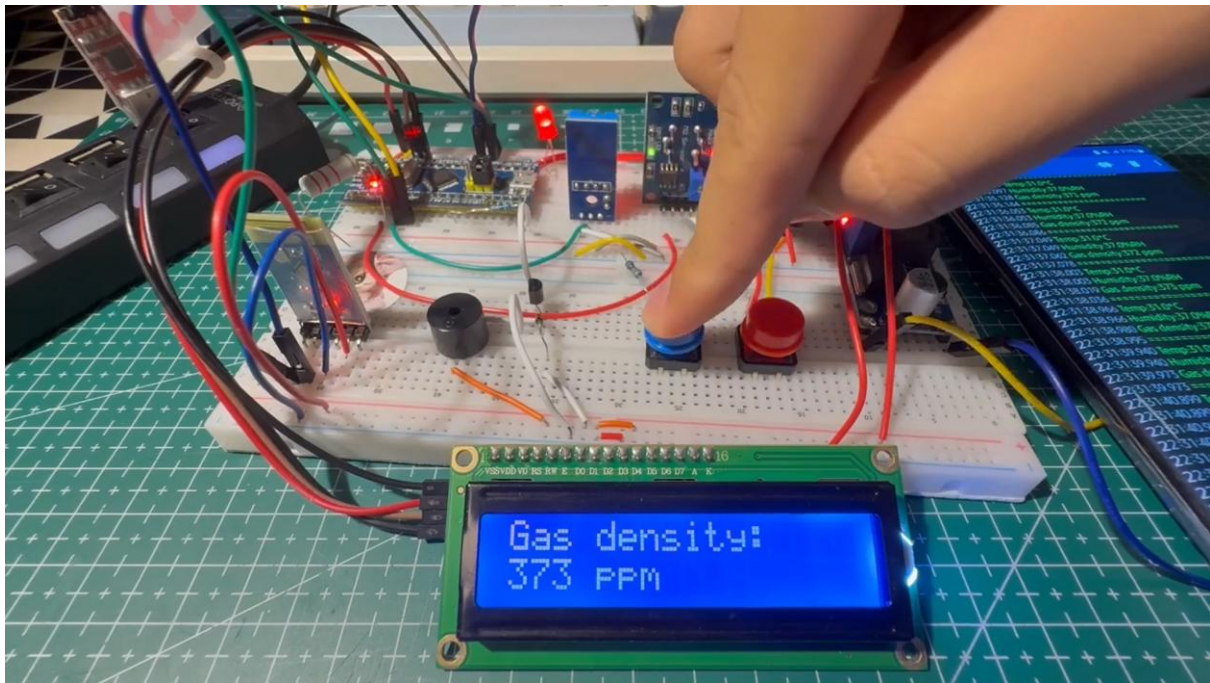


Figure 7.3.1: LCD in displaying gas density mode

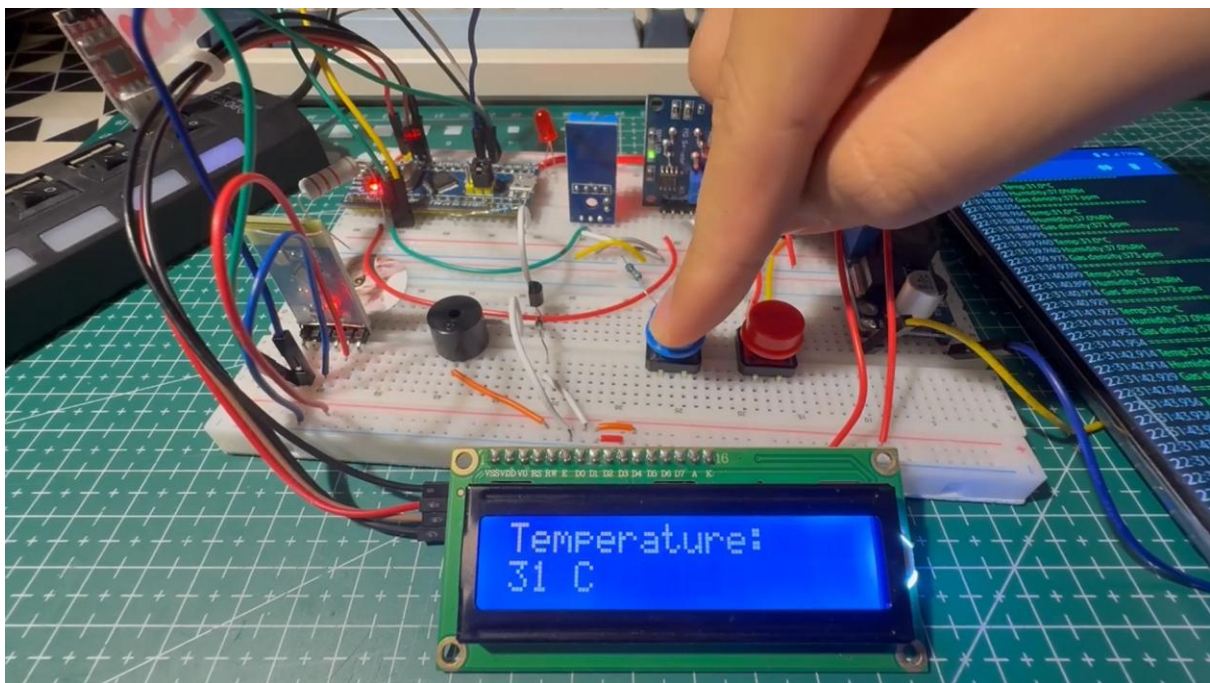


Figure 7.3.2: LCD in displaying temperature mode

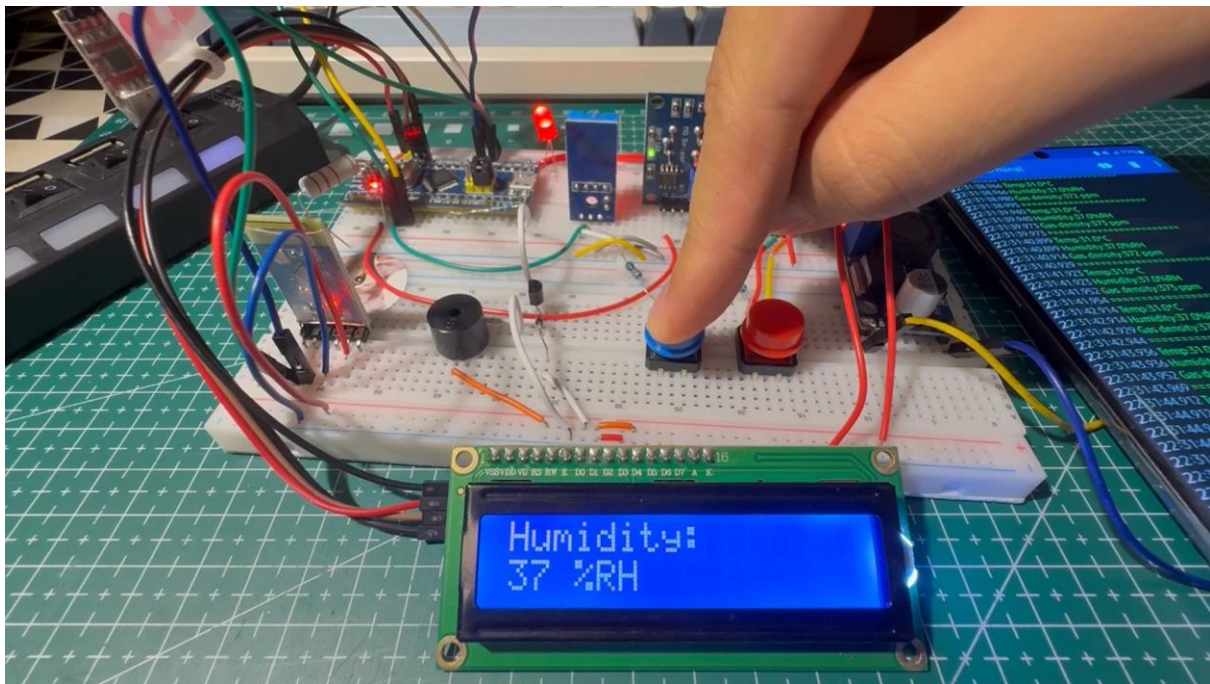


Figure 7.3.3: LCD in displaying humidity mode

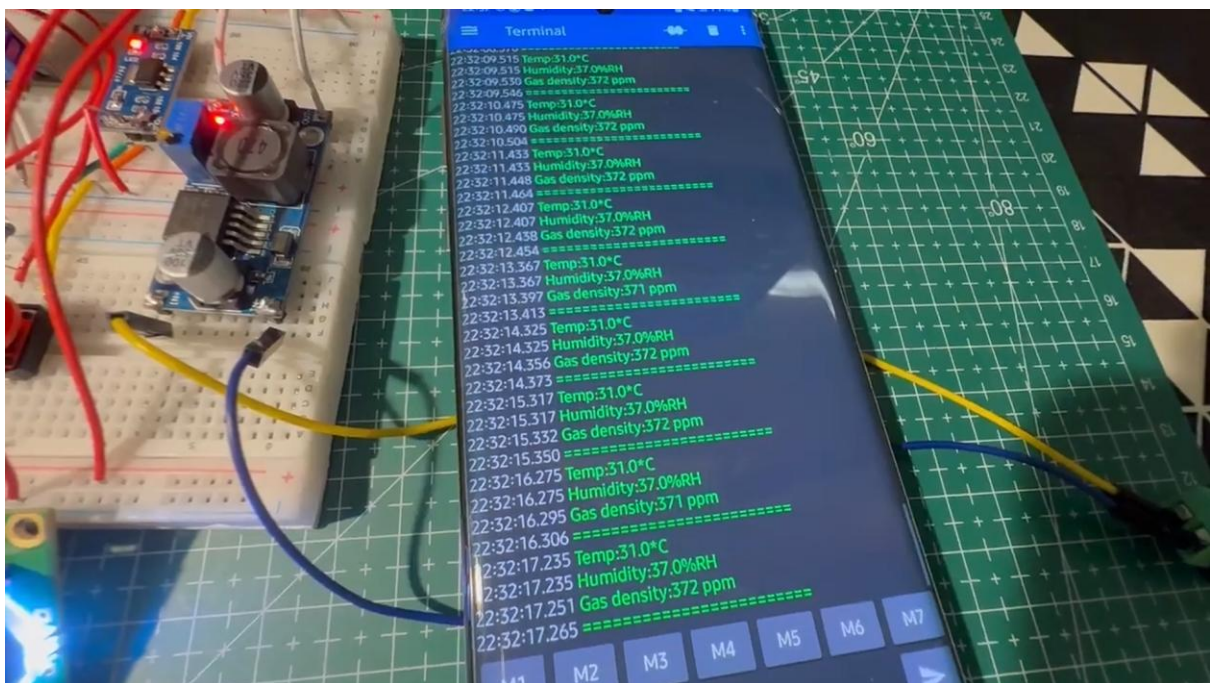


Figure 7.3.4: Phone app successfully received messages from bluetooth module HC-05

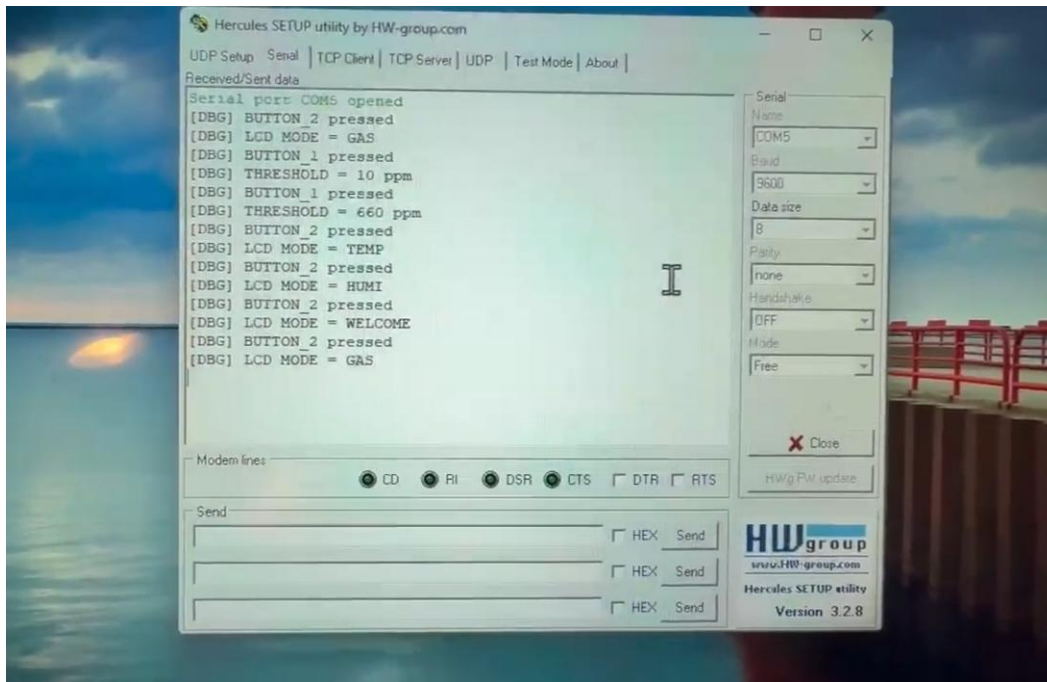


Figure 7.3.5: Hercules successfully received messages from USB to TTL module

Breadboard testing confirmed the final behavior of the prototype on real hardware. The LCD successfully displayed all four modes. The gas value changed according to sensor output and triggered the buzzer according to the selected threshold. BUTTON_1 correctly changed the gas threshold between 660 ppm and 10 ppm, while BUTTON_2 correctly cycled the LCD display modes. Temperature and humidity values were shown locally and also transmitted wirelessly through the HC-05 module to a smartphone serial Bluetooth application.

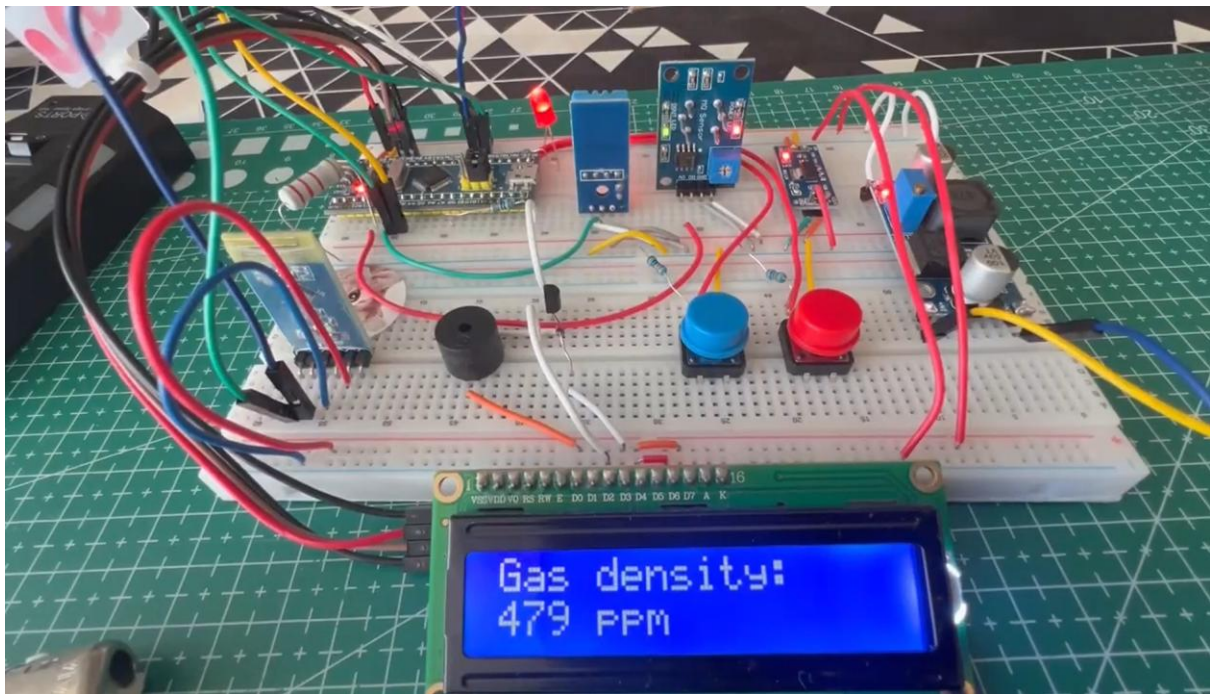


Figure 7.3.6: Read gas density in normal condition

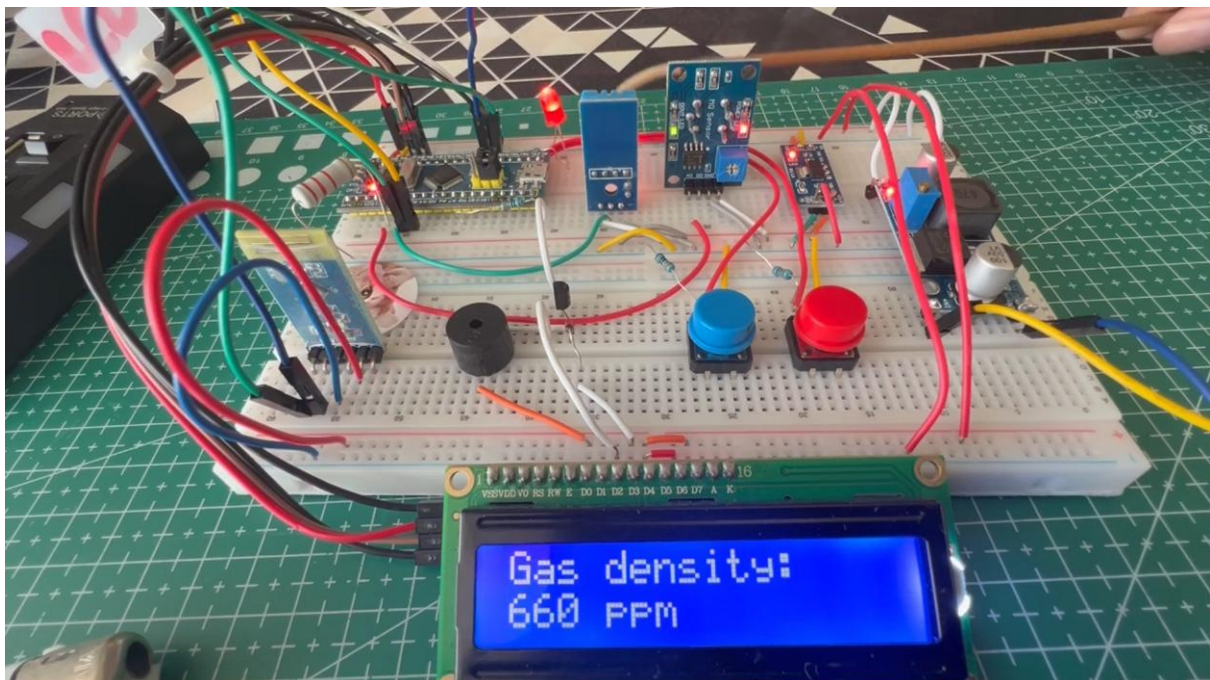


Figure 7.3.7: Read gas density in smoke environment

For the gas sensor test, we burnt an incense stick to generate smoke and put it near the MQ-2 sensor. As observed, the read gas density gradually increased over time until reached 660 ppm, the buzzer was activated. After the incense stick was put away, this number decreased back to its normal condition value and the buzzer was turned off.

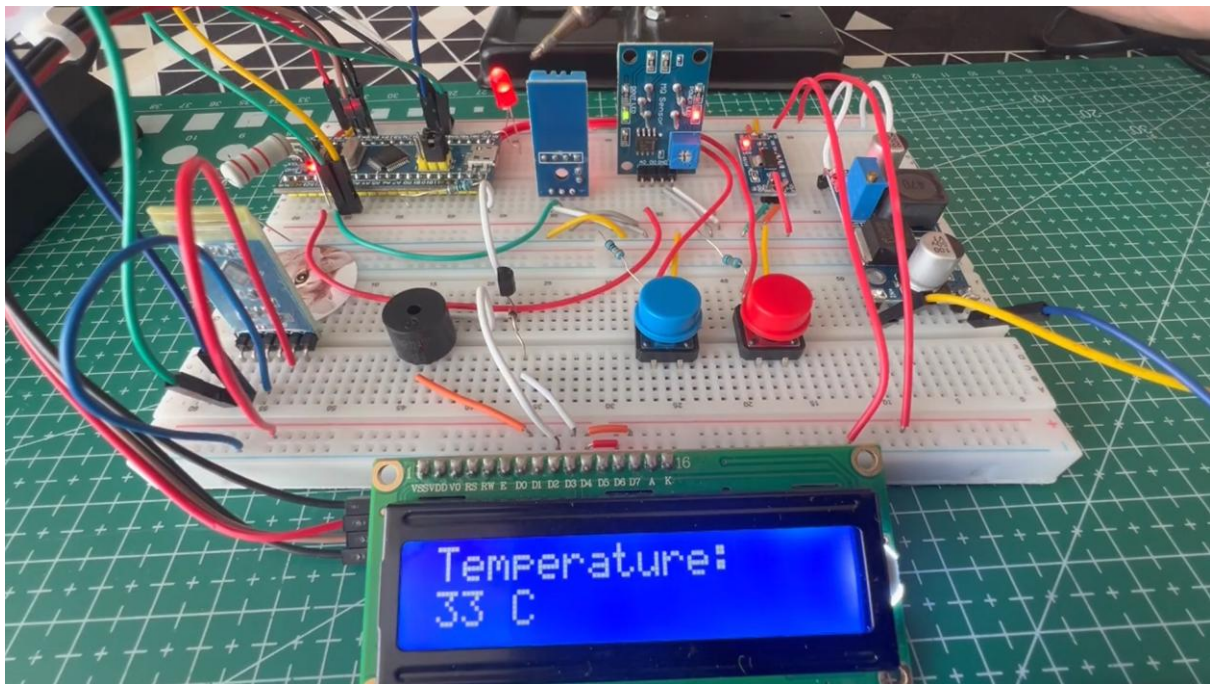


Figure 7.3.8: Read temperature in normal condition

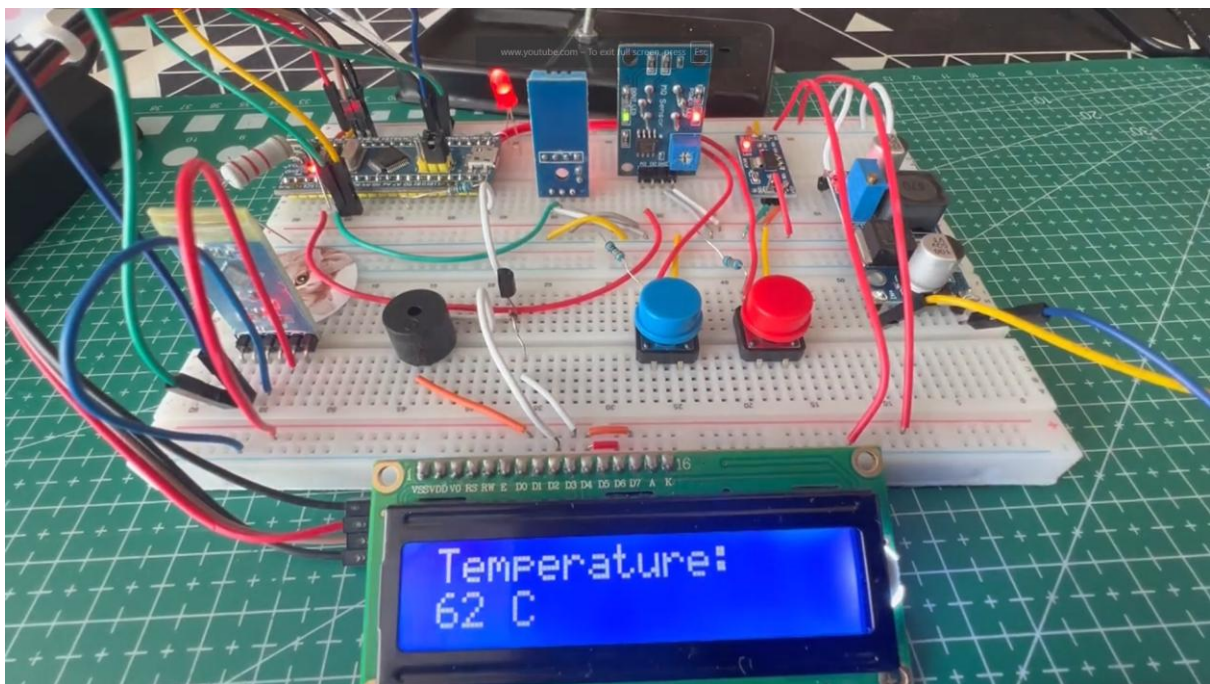


Figure 7.3.9: Read temperature when being near soldering iron

In the temperature test, we use a soldering iron to make heat and put its head next to DHT-11 sensor, the room temperature before the test was 27 Celcius degree, after a period of time, it rocketed to approximately 62 Celcius degree. This shows that the DHT-11 sensor is sensitive to the alternation of temperature.

7.4. PCB bring-up and validation

The PCB was fabricated successfully and basic power and continuity checks were performed. During debugging and firmware refinement, however, the breadboard prototype remained the main platform for subsystem validation because it allowed faster rewiring and easier observation of signals. The PCB therefore demonstrates the manufacturability of the design, while the breadboard demonstrates the final validated functional behavior.

7.5. Requirement verification matrix

Requirement ID	Verification Method	Result
FR-1.1.1: Temperature sensing	Observed on 16x2 LCD (Mode 3) and verified via HC-05 Bluetooth terminal output in °C.	Passed
FR-1.1.2: Humidity sensing	Observed on 16x2 LCD (Mode 4) and verified via HC-05 Bluetooth terminal output in %RH.	Passed
FR-1.1.3: Gas sensing	Monitored approximate ppm-like values on LCD (Mode 2) during exposure to gas sources.	Passed
FR-2.1.1: Threshold switching	Physical interaction with BUTTON_1 to toggle alarm threshold between 660 ppm and 10 ppm.	Passed
FR-2.1.2: LCD mode cycle	Physical interaction with BUTTON_2 to cycle through the four defined display screens.	Passed
FR-3.1: Buzzer alarm	Verified audible activation when gas density exceeded the selected threshold.	Passed

Requirement ID	Verification Method	Result
FR-4.1: Wireless transmission	Verified reception of three separate formatted messages on a mobile Bluetooth serial application.	Passed

7.6. Discussion of results

The final prototype successfully demonstrates the complete embedded workflow from sensing to processing, local display, alarm generation, and wireless reporting. The BSP–APP software structure improved maintainability and made hardware integration more systematic. The strongest results were obtained on breadboard hardware, where LCD operation and wireless data transmission were verified successfully. Some limitations remain: the gas value is an approximate ppm-like scaled value rather than an industrially calibrated gas concentration, and some Proteus peripheral models were not fully reliable. Nevertheless, the final prototype meets the main educational and functional objectives of the project.

The test videos can be view in the following links:

Full test: <https://youtu.be/GwvQwB3FSAY>

Gas test: <https://youtu.be/L1KeKjgLLEk>

Temp test: <https://youtu.be/eYzAX3KRrvM>

8. Conclusion

This project successfully developed a mini environmental station based on the STM32 platform for indoor monitoring applications. The final prototype integrates temperature-humidity sensing, gas sensing, local LCD visualization, audible gas-threshold warning, and wireless data transmission through a Bluetooth UART module. The implementation also demonstrates a modular firmware structure based on BSP and APP separation, which simplified development and debugging. Although some peripheral models in simulation were not fully stable and the gas density value remains approximate rather than industrially calibrated, the real hardware prototype achieved the intended functionality and provided a complete end-to-end embedded system demonstration. Future work may include improved gas calibration, enclosure design, cloud connectivity, and a more accurate environmental sensor set.

9. Reference

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